TRIPLE QUANTUM SCHUBERT CALCULUS

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Abstract.

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1. Introduction

2. Definitions and statement of results

2.1. Quantum double Schubert polynomials and triple quantum Schubert structure constants.

We fix $n \geq 1$ and write

$$x = (x_1, \ldots, x_n), \qquad y = (y_1, \ldots, y_n), \qquad q = (q_1, \ldots, q_{n-1}).$$

For $1 \leq a < b \leq n$, set

$$q_{ab} := q_a q_{a+1} \cdots q_{b-1},$$

and use the cohomological grading $\deg q_i = 2$ (equivalently $\deg q_{ab} = 2(b-a)$).

Let s_i act on the y-variables by swapping y_i, y_{i+1}, and define the divided difference operators

$$\partial_i^{(y)} f := \frac{f - s_i f}{y_i - y_{i+1}} \qquad (1 \leq i \leq n-1).$$

For $w \in S_n$, if $w = s_{i_1} \cdots s_{i_r}$ is any reduced word, set

$$\partial_w^{(y)} := \partial_{i_1}^{(y)} \cdots \partial_{i_r}^{(y)}.$$

Following Kirillov–Maeno [4], define the quantum elementary polynomials $e_j^{(q)}(x_1, \ldots, x_k)$ by

$$\Delta_k(t \mid x_1, \ldots, x_k) := \sum_{j=0}^{k} t^{k-j} e_j^{(q)}(x_1, \ldots, x_k),$$

where

$$\Delta_k(t \mid x_1, \ldots, x_k) = \det \begin{pmatrix} x_1 + t & q_1 & 0 & \cdots & 0 \\ -1 & x_2 + t & q_2 & \ddots & \vdots \\ 0 & -1 & x_3 + t & \ddots & 0 \\ \vdots & \ddots & \ddots & \ddots & q_{k-1} \\ 0 & \cdots & 0 & -1 & x_k + t \end{pmatrix}.$$

Let $w_0 \in S_n$ be the longest permutation. The top quantum double Schubert polynomial is

$$\widetilde{\mathfrak{S}}_{w_0}(x; y) := \prod_{i=1}^{n-1} \Delta_i(-y_{n-i} \mid x_1, \ldots, x_i),$$

and for $w \in S_n$ we define

$$\widetilde{\mathfrak{S}}_w(x; y) := (-1)^{\ell(w_0) - \ell(w)} \partial_{ww_0}^{(y)} \widetilde{\mathfrak{S}}_{w_0}(x; y).$$

Its specialization

$$\widetilde{\mathfrak{S}}_w(x) := \widetilde{\mathfrak{S}}_w(x; 0)$$

is the quantum Schubert polynomial. In particular, at $q = 0$ these recover the ordinary double and single Schubert polynomials.

Given their stability property, we may let the number of variables be infinite, indexing the polynomials by S_∞. As a module over $\mathbb{Z}[q][y]$, the quantum double Schubert polynomials form a basis of $\mathbb{Z}[q][x, y]$, and we may write

$$\widetilde{\mathfrak{S}}_u(x; y)\,\widetilde{\mathfrak{S}}_v(x; z) = \sum_{w,d} q^d c_{u,v}^{w,d}(y; z)\,\widetilde{\mathfrak{S}}_w(x; y)$$

where $d = (d_1, \ldots, d_{n-1})$ is a weak composition and $q^d = q_1^{d_1} \cdots q_{n-1}^{d_{n-1}}$. The coefficients $c_{u,v}^{w,d}(y; z)$ are the *triple quantum Schubert structure constants*, which are polynomials in y and z with integer coefficients. This product respects the grading where for each i, $\deg y_i = \deg z_i = 1$ and $\deg q_i = 2$. In particular, if $c_{u,v}^{w,d}(y; z) \neq 0$, then $\deg c_{u,v}^{w,d}(y; z) = \ell(u) + \ell(v) - \ell(w) + 2\sum_i d_i$. Unlike the usual triple Schubert structure constants, they do not exhibit triangularity in Bruhat order, and indeed we may have $\ell(w) < \ell(u) + \ell(v)$. There is an ordering in which the quantum double Schubert polynomials are triangular, but we will not need this in full generality.

Remark 2.1 ($\overset{k}{\rightsquigarrow}$ and the quantum k-Bruhat order). qrtere

For $d = (0, \ldots, 0)$, this is the ordinary triple Schubert calculus structure constant, and it was proved in [2] that these have the positivity property that they are polynomials in the differences $y_i - z_j$ with nonnegative integer coefficients. We will refer to this as *triple Schubert positivity*. We have the following conjecture:

Theorem 2.2 (Triple quantum Schubert positivity)**.** *For all u, v, w and d, the structure constants $c_{u,v}^{w,d}(y; z)$ are polynomials in the differences $y_i - z_j$ with nonnegative integer coefficients.*

Proof. The proof below is reproduced, with notation adapted to this paper, from the manuscript of Tongren Xiao and Yihua Yue [9], with their permission. Their product is written with the output basis carrying the parameters of the second factor. Commutativity of multiplication therefore identifies their coefficient $C_{v,u}^{w,d}(z, y)$ with $c_{u,v}^{w,d}(y; z)$. It is enough to prove that $C_{u,v}^{w,d}(y, t)$ is in $\mathbb{Z}_{\geq 0}[t_i - y_j]$.

Fix $u, v, w \in S_n$, increasing n if necessary so that all permutations and parameters occurring in the coefficient under consideration are included. Write $\mathbf{a} = (t_1, \ldots, t_n, y_1, \ldots, y_n)$, and let $\tau \in S_{2n}$ interchange the two blocks, so that $\tau\mathbf{a} = (y_1, \ldots, y_n, t_1, \ldots, t_n)$. Regard the product expansion in S_{2n}:

$$\widetilde{\mathfrak{S}}_u(x; \tau\mathbf{a})\,\widetilde{\mathfrak{S}}_v(x; \mathbf{a}) = \sum_{w \in S_{2n}} C_{u,v}^{w,\tau}(\mathbf{a}; q)\,\widetilde{\mathfrak{S}}_w(x; \mathbf{a}). \tag{2.1}$$

We recall the geometric interpretation of these coefficients. Let $e_i(x)$ and $E_i(x, q)$ denote, respectively, the elementary symmetric polynomials and their quantum analogues in $2n$ variables, and set

$$J_{\mathbf{a}}^{2n} = \langle e_i(x) - e_i(\mathbf{a}) : 1 \leq i \leq 2n \rangle, \qquad J_{q\mathbf{a}}^{2n} = \langle E_i(x, q) - e_i(\mathbf{a}) : 1 \leq i \leq 2n \rangle.$$

The Fomin–Gelfand–Postnikov deformation map, extended $\mathbb{Z}[\mathbf{a}]$-linearly, induces

$$\bar{\theta} : \mathbb{Z}[x, \mathbf{a}]/J_{\mathbf{a}}^{2n} \longrightarrow \mathbb{Z}[x, \mathbf{a}, q]/J_{q\mathbf{a}}^{2n}.$$

The classical Borel presentation identifies the source with $H_T^*(X)$, and Kim's presentation identifies the target with $QH_T^*(X)$, where $X = \mathrm{Fl}_{2n}$. Under the Lam–Shimozono identification of quantum double Schubert polynomials with quantum Schubert classes, (2.1) is the quantum equivariant Littlewood–Richardson expansion for the classes obtained by applying $\bar{\theta}$ to the first factor. Consequently,

$$C_{u,v}^{w,\tau}(\mathbf{a}; q) = \sum_{d \geq 0} q^d (\pi_{M_d})_*^T \left(\mathrm{ev}_1^*[\tau X_u]^T \ \mathrm{ev}_2^*[X_v]^T \ \mathrm{ev}_3^*[Y_w]^T \right). \tag{2.2}$$

Here $M_d = \overline{M}_{0,3}(X, d)$, $\mathrm{ev}_i : M_d \to X$ are the evaluation maps, $\pi_{M_d} : M_d \to \mathrm{pt}$ is the structure map, and X_r and Y_r denote the Schubert and opposite Schubert varieties indexed by r.

For each effective degree d, put

$$F = (\mathrm{ev}_1, \mathrm{ev}_2) : M_d \longrightarrow X \times X, \qquad I_{u,v}^d = F^{-1}(\tau X_u \times X_v), \qquad \Gamma_{u,v}^d = (\mathrm{ev}_3)_*[I_{u,v}^d].$$

The incidence scheme $I_{u,v}^d$ is reduced and locally irreducible, and all its irreducible components have codimension $\ell(u) + \ell(v)$. Indeed, the moduli space M_d is irreducible by Thomsen's theorem, and Mihalcea's

incidence lemma applies after diagonal translation by $a_0 = 1 \times w_0 \in S_{2n}$, where w_0 is the longest element in the second block: it identifies $\tau X_u \times X_v$ with $w_0^{(2n)} X_u \times X_v$. The codimension is therefore

$$\operatorname{codim}_X(w_0^{(2n)} X_u) + \ell(v) = \ell(u) + \ell(v).$$

By the incidence formula of Mihalcea, this gives

$$\operatorname{ev}_1^*[\tau X_u]^T \operatorname{ev}_2^*[X_v]^T \cap [M_d]^T = [I^d_{u,v}]_T. \tag{2.3}$$

The cycle $\Gamma^d_{u,v}$ is effective and $B^-(\tau)$-invariant. To see invariance, write $N^-(\tau) = N^- \cap \tau N^- \tau^{-1}$ and $B^-(\tau) = TN^-(\tau)$. The varieties X_v and τX_u are stable under the corresponding actions, so their product, its inverse image under the equivariant map F, and its proper push-forward under ev_3 are $B^-(\tau)$-invariant. Effectivity follows from the reducedness of $I^d_{u,v}$.

There is a unique expansion

$$[\Gamma^d_{u,v}]_T = \sum_{r \in S_{2n}} p^{r,d}_{u,v}(\mathbf{a})[X_r]_T \quad \text{in } H^*_T(X).$$

By the Graham–positivity theorem for $B^-(\tau)$-invariant effective cycles,

$$p^{r,d}_{u,v}(\mathbf{a}) \in \mathbb{Z}_{\geq 0}[t_i - y_j : 1 \leq i, j \leq n].$$

Finally, by equivariant Poincaré duality, (2.3), and the projection formula,

$$(\operatorname{ev}_3)^T_* \left(\operatorname{ev}_1^*[\tau X_u]^T \operatorname{ev}_2^*[X_v]^T \right) = [\Gamma^d_{u,v}]_T,$$

and hence the degree-d summand of (2.2) is

$$(\pi_X)^T_* \left([\Gamma^d_{u,v}]_T [Y_w]^T \right) = \sum_{r \in S_{2n}} p^{r,d}_{u,v}(\mathbf{a}) (\pi_X)^T_* \left([X_r]^T [Y_w]^T \right) = p^{w,d}_{u,v}(\mathbf{a}).$$

Thus every coefficient $C^{w,d}_{u,v}(y,t)$ is in $\mathbb{Z}_{\geq 0}[t_i - y_j]$. Replacing (u,v,y,t) by (v,u,z,y) proves the asserted positivity of $c^{w,d}_{u,v}(y;z)$. □

Example 2.3. We compute the product $\widetilde{\mathfrak{S}}_{15423}(x;y)\,\widetilde{\mathfrak{S}}_{1432}(x;z)$:

$$\begin{aligned}
&q_2q_3\,(y_1+y_4-z_1-z_2)\,\widetilde{\mathfrak{S}}_{124635}\,(x;y) + q_2q_3\,(y_1+y_4-z_1-z_2)\,\widetilde{\mathfrak{S}}_{13452}\,(x;y)\\
&+ q_2q_3\,(y_1+y_4-z_1-z_2)\,\widetilde{\mathfrak{S}}_{14253}\,(x;y)\\
&+ q_2q_3\,\widetilde{\mathfrak{S}}_{125634}\,(x;y) + q_2q_3\,\widetilde{\mathfrak{S}}_{126435}\,(x;y) + q_2q_3\,\widetilde{\mathfrak{S}}_{13542}\,(x;y) + q_2q_3\,\widetilde{\mathfrak{S}}_{214635}\,(x;y)\\
&+ q_2q_3\,\widetilde{\mathfrak{S}}_{23451}\,(x;y) + q_2q_3\,\widetilde{\mathfrak{S}}_{24153}\,(x;y) + q_2q_3\,\widetilde{\mathfrak{S}}_{31452}\,(x;y) + q_2q_3\,\widetilde{\mathfrak{S}}_{41253}\,(x;y)\\
&+ q_2\,(y_1+y_4-z_1-z_2)\,\widetilde{\mathfrak{S}}_{146235}\,(x;y) + q_2\,(y_4+y_5-z_1-z_2)\,\widetilde{\mathfrak{S}}_{24513}\,(x;y)\\
&+ q_2\,\widetilde{\mathfrak{S}}_{246135}\,(x;y) + q_3\,\widetilde{\mathfrak{S}}_{1724356}\,(x;y) + q_3\,\widetilde{\mathfrak{S}}_{261435}\,(x;y) + q_3\,\widetilde{\mathfrak{S}}_{35142}\,(x;y)\\
&+ (q_2q_3 + q_2\,((y_1-z_2)\,(y_4+y_5-z_1-z_2) + (y_4-z_1)\,(y_5-z_1)))\,\widetilde{\mathfrak{S}}_{14523}\,(x;y)\\
&+ ((y_4-z_1)\,(y_5-z_3) + (y_4-z_1)\,(y_1+y_2-z_1-z_2)\\
&\quad + (y_5-z_2)\,(y_5-z_3) + (y_5-z_2)\,(y_1+y_2-z_1-z_2))\,\widetilde{\mathfrak{S}}_{25413}\,(x;y)\\
&+ ((y_1-z_2)\,(y_1-z_3)\,(y_4-z_2) + (y_1-z_2)\,(y_1-z_3)\,(y_5-z_1) + (y_1-z_2)\,(y_5-z_1)\,(y_5-z_2)\\
&\quad + (y_1-z_3)\,(y_4-z_1)\,(y_5-z_1) + (y_4-z_1)\,(y_5-z_1)\,(y_5-z_2))\,\widetilde{\mathfrak{S}}_{15423}\,(x;y)\\
&+ (q_2 + (y_1-z_2)\,(y_1-z_3) + (y_1-z_3)\,(y_5-z_1) + (y_5-z_1)\,(y_5-z_2))\,\widetilde{\mathfrak{S}}_{156234}\,(x;y)\\
&+ (q_3 + (y_1-z_1)\,(y_4-z_1) + (y_5-z_2)\,(y_1+y_4-z_1-z_2))\,\widetilde{\mathfrak{S}}_{45123}\,(x;y)\\
&+ (q_2q_3 + q_3\,((y_1-z_1)\,(y_1+y_5-z_2-z_3) + (y_5-z_2)\,(y_5-z_3)))\,\widetilde{\mathfrak{S}}_{15243}\,(x;y)\\
&+ q_3\,(y_1+y_5+y_6-z_1-z_2-z_3)\,\widetilde{\mathfrak{S}}_{162435}\,(x;y) + q_3\,(y_1+y_2+y_5-z_1-z_2-z_3)\,\widetilde{\mathfrak{S}}_{25143}\,(x;y)\\
&+ q_2q_3\,(y_1+y_2+y_5-z_1-z_2-z_3)\,\widetilde{\mathfrak{S}}_{12543}\,(x;y) + q_2q_3\,(y_1+y_2+y_5-z_1-z_2-z_3)\,\widetilde{\mathfrak{S}}_{21453}\,(x;y)\\
&+ q_2q_3\,((y_1-z_2)(y_1-z_3) + (y_1-z_2)(y_2-z_2) + (y_1-z_2)(y_4-z_1)\\
&\quad + (y_1-z_2)(y_5-z_1) + (y_2-z_1)(y_4-z_1) + (y_4-z_1)(y_5-z_3))\,\widetilde{\mathfrak{S}}_{12453}\,(x;y)
\end{aligned}$$

$$
\begin{aligned}
&+ (y_1 + y_4 - z_1 - z_2)\,\widetilde{\mathfrak{S}}_{1742356}(x;y) + (y_1 + y_4 - z_1 - z_2)\,\widetilde{\mathfrak{S}}_{461235}(x;y)\\
&+ (y_4 + y_5 - z_1 - z_2)\,\widetilde{\mathfrak{S}}_{35412}(x;y) + (y_4 + y_5 - z_1 - z_2)\,\widetilde{\mathfrak{S}}_{45213}(x;y)\\
&+ ((y_1 - z_2)(y_1 - z_3) + (y_1 - z_2)(y_5 + y_6 - z_1 - z_2)\\
&\quad + (y_1 - z_3)(y_4 - z_1) + (y_4 - z_1)(y_5 + y_6 - z_1 - z_2))\,\widetilde{\mathfrak{S}}_{164235}(x;y)\\
&+ (y_1 + y_2 + y_5 - z_1 - z_2 - z_3)\,\widetilde{\mathfrak{S}}_{256134}(x;y) + (y_1 + y_5 + y_6 - z_1 - z_2 - z_3)\,\widetilde{\mathfrak{S}}_{165234}(x;y)\\
&+ (y_1 + y_2 + y_4 + y_5 + y_6 - 2z_1 - 2z_2 - z_3)\,\widetilde{\mathfrak{S}}_{264135}(x;y)\\
&+ \widetilde{\mathfrak{S}}_{1752346}(x;y) + \widetilde{\mathfrak{S}}_{265134}(x;y) + \widetilde{\mathfrak{S}}_{2741356}(x;y)\\
&+ \widetilde{\mathfrak{S}}_{356124}(x;y) + \widetilde{\mathfrak{S}}_{364125}(x;y) + \widetilde{\mathfrak{S}}_{462135}(x;y)\\
&+ \widetilde{\mathfrak{S}}_{561234}(x;y)
\end{aligned}
$$

By inspection, we can see that the coefficients exhibit triple Schubert poisitivity.

We will provide formulas below that demonstrate this in special cases. In particular, we prove

Theorem 2.4. *Suppose $u, v, w \in S_\infty$ and v is k-Grassmannian for some k. Then for any d, $c_{u,v}^{w;d}(y;z)$ exhibits triple Schubert positivity.*

2.2. **The parabolic case.** Unlike ordinary cohomology or equivariant cohomology, quantum cohomology and equivariant quantum cohomology do not demonstrate simple functoriality, in the sense that a projection of the complete flag variety G/B to a partial flag variety G/P does not induce a homomorphism of rings $QH_T^*(G/P) \to QH_T^*(G/B)$, and indeed the quantum cohomology of G/P is not isomorphic to the subring of the quantum cohomology of G/B generated by the Schubert classes indexed by W^P. This is reflected in the fact that there are parabolic quantum double Schubert polynomials defined in a different manner that represent Schubert classes in $QH_T^*(G/P)$ [5] and are not in general equal to the quantum double Schubert polynomials above.

Definition 2.5 (Parabolic quantum double Schubert polynomials)**.** Following Lam and Shimozono [5], we define parabolic quantum double Schubert polynomials as follows. Fix an integer $n > 0$. A parabolic subgroup W_P of S_n, which has Coxeter generators $s_1, \ldots, s_{n-1}$, is generated by a subset $P \subseteq \{s_1, \ldots, s_{n-1}\}$. In terms of the subgroup's properties, it is most natural to identify P by the reflections that it *excludes*. If P is generated by $S \setminus \{s_{i_1}, \ldots, s_{i_k}\}$ in increasing order, we define integers $n_1, \ldots, n_{k+1}$ with

$$n_1 + \cdots + n_{k+1} = n$$

by $n_1 = i_1$ and $n_p = i_p - i_{p-1}$ for $2 \le p \le k+1$. Then N_j for $j \ge 1$ is defined by

$$N_j = n_1 + \cdots + n_j$$

for $0 \le j \le k+1$.

For such a setup, define a matrix D_j^P to be the $n \times n$ matrix with $x_1, \ldots, x_n$ on the diagonal, -1 on the superdiagonal, and $(-1)^{n_{j+1}+1} q_j$ at $(N_{j+1}, N_{j-1}+1)$ for $1 \le j \le k$. Notice that there are precisely k q variables, and we set

$$\deg q_j = n_j + n_{j+1}$$

Define

$$\Delta_j^P(t;x) = \det(D_j^P - t\mathrm{Id})$$

Then we define the parabolic quantum double Schubert polynomials $\widetilde{\mathfrak{S}}_w^P(x;y)$ for $w \in W^P$ by

$$\widetilde{\mathfrak{S}}_w^P(x;y) = (-1)^{\ell(w(w_0^P)^{-1})} \partial^{(y)}_{w(w_0^P)^{-1}} \prod_{j=1}^{k} \Delta_j^P(-y_{n-N_{j+1}+1}, \ldots, -y_{n-N_j}; x)$$

where w_0^P is the longest element of W^P.

Taken as defined, when we vary n these polynomials are not stable under inclusion and do not form a basis of $\mathbb{Z}[q][x_1, \ldots, x_n; y]$ as a module over $\mathbb{Z}[q][y]$. To obtain stability and a basis, instead we fix n and choose $N > n$ as large as we would like, maintaining $N_{k+1} = n$ and setting $n_{k+2} = N - n$. From a cohomological

perspective, this introduces "extraneous" polynomials that are indexed by elements not in W^P. The full indexing set is $W^{\widehat{P}} \subset S_\infty$, where

$$\widehat{P} = P \cup \{s_{n+1}, s_{n+2}, \ldots\}$$

Example 2.6. As equivariant cohomology rings of finite dimensional projective varieties over $\mathbb{C}$ are finitely generated modules over the coefficient ring, it is fully expected that products of double Schubert polynomials may include terms indexed by permutations that do not lie within the indexing set of the basis of Schubert classes. For example, consider the following non-quantum double Schubert polynomial product, with $P = \{s_1, s_3\}$ and basis elements in S_4:

$$\begin{aligned}\mathfrak{S}_{2413}(x;y)\,\mathfrak{S}_{3412}(x;y) = &-(\alpha_1+\alpha_2)(\alpha_1+\alpha_2+\alpha_3)(\alpha_2+\alpha_3)\ \mathfrak{S}_{3412}(x;y)\\ &+(\alpha_1+\alpha_2)(\alpha_1+2\alpha_2+2\alpha_3+\alpha_4)\ \mathfrak{S}_{35124}(x;y)\\ &-(\alpha_1+\alpha_2)\ \mathfrak{S}_{361245}(x;y)\\ &-(\alpha_1+2\alpha_2+2\alpha_3+\alpha_4)\ \mathfrak{S}_{45123}(x;y)\\ &+\mathfrak{S}_{461235}(x;y).\end{aligned}$$

Not only does the product $\widetilde{\mathfrak{S}}^P_{2413}(x;y)\,\widetilde{\mathfrak{S}}^P_{3412}(x;y)$ include extra terms indexed by permutations in W^P, there are also terms with nonzero coefficients that are in $W^{\widehat{P}} \setminus W^P$:

$$\begin{aligned}q_1(\alpha_1\alpha_2+\alpha_1\alpha_3+\alpha_2\alpha_3+\alpha_2^2)\ \widetilde{\mathfrak{S}}^P_{132}(x;y) &+ q_1(\alpha_1+\alpha_2)\ \widetilde{\mathfrak{S}}^P_{231}(x;y)\\ &+ q_1(\alpha_2+\alpha_3)\ \widetilde{\mathfrak{S}}^P_{1423}(x;y)\\ &+ q_1(\alpha_1+\alpha_2)\ \widetilde{\mathfrak{S}}^{\widehat{P}}_{13254}(x;y)\\ &+ q_1\ \widetilde{\mathfrak{S}}^{\widehat{P}}_{14253}(x;y)\\ &+ q_1\ \widetilde{\mathfrak{S}}^P_{15234}(x;y)\\ &+ q_1\ \widetilde{\mathfrak{S}}^P_{2413}(x;y)\end{aligned}$$

Definition 2.7 (q-vectors and Peterson-Woodward lifts)**.** Define $Q^\vee$, for legacy purposes, to be the set of weak compositions. For an integer i and $d \in Q^\vee$, define

$$\langle d\rangle_i = 2d_i - d_{i-1} - d_{i+1}$$

The double indexed version will be denoted by

$$\langle d\rangle_{ij} = \sum_{k=i}^{j-1} \langle d\rangle_k$$

For a parabolic subgroup W_P, we define $Q^\vee_P$ to be the set of weak compositions d such that $\langle d\rangle_i = 0$ for all $s_i \notin P$. The q-vectors in W^P are indexed by $Q^\vee/Q^\vee_P$.

Let $d_P \in Q^\vee/Q^\vee_P$. A *Peterson-Woodward lift* of d_P is a $d \in Q^\vee$ such that $d_P \in d + Q^\vee_P$ and $\langle d\rangle_{ij} \in \{0,-1\}$ for all i,j such that $s_i, \ldots, s_{j-1} \in P$.

Method 2.8 (Checking the Peterson-Woodward lift property)**.** While backing out a Peterson-Woodward lift is difficult, going in the other direction is not. Suppose $d \in Q^\vee$ and P is parabolic. If there is any $i \in P$ such that $\langle d\rangle_i \notin \{0,-1\}$, then d is not a Peterson-Woodward lift. If this condition does hold, let $z_P(d)$ be the set of all indexes i such that $\langle d\rangle_i = 0$. This set is naturally divided into maximal disjoint intervals $I_1, \ldots, I_m$. We must check the additional condition that for each such I and each pair $i < j$ with $i, j-1 \in I$, the condition $\langle d\rangle_{ij} \in \{0,-1\}$ holds. If and only if it does, then d is a Peterson-Woodward lift of a unique $d_P \in Q^\vee/Q^\vee_P$. We denote this d_P by $\pi_P(d)$.

This algorithm is polynomial in the length of d, which in almost all currently feasible cases is quite small. In contrast, a naive search for d with d_P as the input has an exponential search space, though a more refined algorithm may exist.

We prove the following result.

Proposition 2.9. *Let W_P be a parabolic subgroup of S_∞ with $P = \widehat{P}$ and let $u, v, w \in W^P$. Then for any $d_P = \pi_P(d) \in Q^\vee/Q_P^\vee$, we have that*

$$c_{u,v}^{w,d_P}(y;z) = c_{u,v}^{ww_P w_{z_P(d)},d}(y;z),$$

Thus, though we lack a simple functoriality property, we can reduce the computation of parabolic triple quantum Schubert structure constants completely to the complete flag variety case with an efficient algorithm. We obtain a nontrivial corollary from this.

Example 2.10. The product in the example above, with $P = \widehat{P}$ satisfying $P \cap \{s_1, s_2, s_3, s_4, s_5\} = \{s_1, s_4\}$, allows us to compute the triple structure constants for the corresponding parabolic quantum double Schubert polynomials:

$$\begin{aligned}
&q_1q_2\,(y_1+y_2+y_5-z_1-z_2-z_3)\,\widetilde{\mathfrak{S}}^P_{1243}\,(x;y)\\
&+q_1q_2\,\widetilde{\mathfrak{S}}^P_{124365}\,(x;y)+q_1q_2\,\widetilde{\mathfrak{S}}^P_{1342}\,(x;y)+q_1q_2\,\widetilde{\mathfrak{S}}^P_{1423}\,(x;y)\\
&+q_2\,((y_1-z_2)\,(y_1-z_3)+(y_1-z_3)\,(y_5-z_1)+(y_5-z_1)\,(y_5-z_2))\,\widetilde{\mathfrak{S}}^P_{15234}\,(x;y)\\
&+q_2\,(y_1+y_5+y_6-z_1-z_2-z_3)\,\widetilde{\mathfrak{S}}^P_{162345}\,(x;y)+q_2\,(y_1+y_2+y_5-z_1-z_2-z_3)\,\widetilde{\mathfrak{S}}^P_{25134}\,(x;y)\\
&+((y_1-z_2)\,(y_1-z_3)\,(y_4-z_2)+(y_1-z_2)\,(y_1-z_3)\,(y_5-z_1)+(y_1-z_2)\,(y_5-z_1)\,(y_5-z_2)\\
&\quad+(y_1-z_3)\,(y_4-z_1)\,(y_5-z_1)+(y_4-z_1)\,(y_5-z_1)\,(y_5-z_2))\,\widetilde{\mathfrak{S}}^P_{15423}\,(x;y)\\
&+q_2\,\widetilde{\mathfrak{S}}^P_{1723456}\,(x;y)+q_2\,\widetilde{\mathfrak{S}}^P_{261345}\,(x;y)+q_2\,\widetilde{\mathfrak{S}}^P_{35124}\,(x;y)\\
&+((y_1-z_2)\,(y_1-z_3)+(y_1-z_3)\,(y_5-z_1)+(y_5-z_1)\,(y_5-z_2))\,\widetilde{\mathfrak{S}}^P_{156234}\,(x;y)\\
&+((y_1-z_2)\,(y_1-z_3)+(y_1-z_2)\,(y_5+y_6-z_1-z_2)\\
&\quad+(y_1-z_3)\,(y_4-z_1)+(y_4-z_1)\,(y_5+y_6-z_1-z_2))\,\widetilde{\mathfrak{S}}^P_{164235}\,(x;y)\\
&+((y_4-z_1)\,(y_5-z_3)+(y_4-z_1)\,(y_1+y_2-z_1-z_2)\\
&\quad+(y_5-z_2)\,(y_5-z_3)+(y_5-z_2)\,(y_1+y_2-z_1-z_2))\,\widetilde{\mathfrak{S}}^P_{25413}\,(x;y)\\
&+((y_1-z_1)\,(y_4-z_1)+(y_5-z_2)\,(y_1+y_4-z_1-z_2))\,\widetilde{\mathfrak{S}}^P_{45123}\,(x;y)\\
&+(y_1+y_5+y_6-z_1-z_2-z_3)\,\widetilde{\mathfrak{S}}^P_{165234}\,(x;y)+(y_1+y_2+y_5-z_1-z_2-z_3)\,\widetilde{\mathfrak{S}}^P_{256134}\,(x;y)\\
&+(y_1+y_4-z_1-z_2)\,\widetilde{\mathfrak{S}}^P_{1742356}\,(x;y)+(y_1+y_4-z_1-z_2)\,\widetilde{\mathfrak{S}}^P_{461235}\,(x;y)\\
&+(y_1+y_2+y_4+y_5+y_6-2z_1-2z_2-z_3)\,\widetilde{\mathfrak{S}}^P_{264135}\,(x;y)\\
&+(y_4+y_5-z_1-z_2)\,\widetilde{\mathfrak{S}}^P_{35412}\,(x;y)+(y_4+y_5-z_1-z_2)\,\widetilde{\mathfrak{S}}^P_{45213}\,(x;y)\\
&+\widetilde{\mathfrak{S}}^P_{1752346}\,(x;y)+\widetilde{\mathfrak{S}}^P_{265134}\,(x;y)+\widetilde{\mathfrak{S}}^P_{2741356}\,(x;y)\\
&+\widetilde{\mathfrak{S}}^P_{356124}\,(x;y)+\widetilde{\mathfrak{S}}^P_{364125}\,(x;y)+\widetilde{\mathfrak{S}}^P_{462135}\,(x;y)\\
&+\widetilde{\mathfrak{S}}^P_{561234}\,(x;y)
\end{aligned}$$

For an illustration, consider $d_P = (-,0,1,-,0)$ for the coefficient of 162345. $q^{d_P} = q_2$. The Peterson-Woodward lift of d_P is $d = (0,0,1,0,0)$, and $w_P = 21354$. $z_P(d) = \{1\}$, so $w_P w_{z_P(d)} = 12354$. Thus, $ww_P w_{z_P(d)} = 162435$, and the q_2-component of the coefficient of $\widetilde{\mathfrak{S}}^P_{162345}(x;y)$ is $(y_1+y_5+y_6-z_1-z_2-z_3)$, which is the same as the q_3-component (q^d) of the coefficient of $\widetilde{\mathfrak{S}}_{162435}(x;y)$ in the complete flag variety case (Example 2.3).

Corollary 2.11 (Triple quantum Schubert positivity for the Grassmannian case)**.** *Suppose $u, v, w \in S_n$ are k-Grassmannian, where P is the corresponding parabolic subroup such that $n_1 = k$ and $n_2 = n-k$. Then $c_{u,v}^{w;d}(y;z)$ exhibits triple Schubert positivity.*

3. Pieri

Definition 3.1 ($\overset{k}{\rightsquigarrow}$, $\mathrm{fix}_k(u,w)$, $q_k(u,w)$, $E_{p,k}(y;z)$, $\mathbf{E}(y_A;z)$)**.** We define a relation $\overset{k}{\rightsquigarrow}$ on pairs of permutations, introduced in its non-quantum form by Sottile [8], which was used originally to indicate when the coefficient of a Schubert polynomial is nonzero in a suitable application of the Pieri formula. In the generalization to double Schubert polynomials in [7], the same relation applies to all nonzero coefficients in the expansion of the generalized Pieri formula. The relation plays the same role in our case.

Given $u, u' \in S_\infty$ and a positive integer k, we declare that $u \overset{k}{\rightsquigarrow} w$ if there exists a p with $0 \leq p \leq k$ such that there are transpositions $t_{a_1b_1}, \ldots, t_{a_pb_p}$ satisfying

(1) $a_i \leq k < b_i$ for all i
(2) $a_i \neq a_j$ whenever $i \neq j$
(3) $b_i \geq b_j$ whenever $i < j$
(4) We have either
$$\ell(ut_{a_1b_1}\cdots t_{a_ib_i}) = \ell(ut_{a_1b_1}\cdots t_{a_{i-1}b_{i-1}}) + 1$$
or
$$\ell(ut_{a_1b_1}\cdots t_{a_ib_i}) = \ell(ut_{a_1b_1}\cdots t_{a_{i-1}b_{i-1}}) - 2(b_i - a_i) + 1$$
for all i
(5) $w = ut_{a_1b_1}\cdots t_{a_pb_p}$

In contrast to the non-quantum case, $u \overset{k}{\rightsquigarrow} w$ does not imply that $\ell(u) \leq \ell(w)$, as the final condition can cause the length of the permutation to drop. For intuition, the condition $\ell(ut_{ab}) = \ell(u) - 2(b-a) + 1$ means that if we write the reflection t_{ab} as
$$s_b s_{b-1} \cdots s_{a+1} s_a s_{a+1} \cdots s_{b-1} s_b$$
then each simple reflection in this word cancels a corresponding simple reflection in a reduced word for u, as opposite to the length-increase case where there is a word for which precisely one simple reflection is inserted, regardless of the length of the reflection as a permutation.

If $u \overset{k}{\rightsquigarrow} w$, define
$$\mathrm{fix}_k(u,w) = \{u(i) \mid i \leq k \text{ and } u(i) = w(i)\}$$
and
$$n_k(u,w) = k - |\mathrm{fix}_k(u,w)|.$$
Also, we define $q_k(u,w)$ recursively. If $u = w$, define $q_k(u,w) = 1$. If we have defined $q_k(u,w')$ with $w' = t_{a_1b_1}\cdots t_{a_{p-1}b_{p-1}}$, suppose $\ell(w't_{a_pb_p}) = \ell(w') + 1$. Then define
$$q_k(u,w) = q_k(u,w')$$
If instead $\ell(w't_{a_pb_p}) = \ell(w') - 2(b_p - a_p) + 1$, define
$$q_k(u,w) = q_k(u,w')q_{a_pb_p}$$

Suppose y and z are sets of variables and $0 \leq p \leq k$ are integers. We define $E_{p,k}(y;z)$ to be the factorial elementary symmetric polynomial in y and z. For specificity, this is the double Schubert polynomial $\mathfrak{S}_{c[p,k]}(y;z)$ where
$$c[p,k] = s_{k-p+1}\cdots s_{k-1}s_k$$
If $p = 0$, the polynomial is identically 1. If $p = k$, $E_{k,k}(y;z)$ can be simply written as
$$E_{k,k}(y;z) = \prod_{i=1}^{k}(y_i - z_1)$$
An explicit formula can be written for general p without too much trouble as well, using e.g. [6]:
$$E_{p,k}(y;z) = \sum_{1\leq i_1<\cdots<i_p\leq k}\prod_{j=1}^{p}(y_{i_j} - z_{i_j-j+1})$$

For a set $A = \{i_1, \ldots, i_{|A|}\}$, we define
$$\mathbf{E}(y_A;z) = \prod_{i\in A}(y_i - z_1) = E_{|A|,|A|}(y_{i_1}, \ldots, y_{i_{|A|}};z),$$

the specialization of the factorial elementary polynomial to which the coefficients of the Pieri formula reduce. It will be used to express those coefficients.

Theorem 3.2 (The Pieri formula)**.** *Suppose $u \in S_\infty$ and $k \geq 1$. Then*

$$\widetilde{\mathfrak{S}}_u(x;y)\widetilde{E}_{k,k}(x;z) = \sum_{u \overset{k}{\rightsquigarrow} w} q_k(u,w)\mathbf{E}(y_{\mathrm{fix}_k(u,w)};z)\,\widetilde{\mathfrak{S}}_w(x;y)$$

4. Algorithms for multiplying (quantum) double Schubert polynomials

Definition 4.1 ($\mathfrak{c}$, (strict) dominant permutations, θ, μ, $\widetilde{\theta}$, $\widetilde{\mu}$, medium theta $\overline{\theta}$, $\overline{\mu}$)**.** We say that a permutation μ is *dominant* if $\mathfrak{c}_i(\mu) \geq \mathfrak{c}_{i+1}(\mu)$ for all i. A permutation μ is said to be *strict dominant* if μ is dominant and whenever $\mathfrak{c}_i(\mu) \neq 0$ we have $\mathfrak{c}_i(\mu) > \mathfrak{c}_{i+1}(\mu)$. For a permutation $v \in S_\infty$, we define a dominant permutation $\mu(v)$ recursively as follows. If v is already dominant, define $\mu(v) = v$. Otherwise, let i be the maximal index such that $\mathfrak{c}_i(v) < \mathfrak{c}_{i+1}(v)$, and define

$$\mu(v) = \mu(vs_i)$$

Define $\theta_i(v)$ to be $\mathfrak{c}_i(\mu(v))$.

Similarly, for a permutation $v \in S_\infty$, we define a strict dominant permutation $\widetilde{\mu}(v)$ recursively as follows. If v is already strict dominant, define $\widetilde{\mu}(v) = v$. Otherwise, let i be the maximal index such that $\mathfrak{c}_{i+1}(v) \neq 0$ and $\mathfrak{c}_i(v) \leq \mathfrak{c}_{i+1}(v)$, then define

$$\widetilde{\mu}(v) = \widetilde{\mu}(vs_i)$$

Define $\widetilde{\theta}_i(v)$ to be $\mathfrak{c}_i(\widetilde{\mu}(v))$.

Between these two lies the *medium theta* $\overline{\theta}(v)$, the partition actually iterated by the quantum multiplication algorithm. Like $\theta(v)$ it is a weakly decreasing partition obtained from $\mathfrak{c}(v)$ by incrementing and reordering, but it stops short of separating every pair of equal parts: a pair of equal consecutive nonzero parts $\overline{\theta}_i(v) = \overline{\theta}_{i+1}(v)$ is left in place exactly when it is preceded by a strict drop of at least two, that is $\overline{\theta}_{i-1}(v) \geq \overline{\theta}_i(v) + 2$ (or $i = 1$), and every other ascent or equality is removed by incrementing as for θ. Thus $\overline{\theta}(v)$ permits some, but not all, repeated parts, and lies between $\theta(v)$ (arbitrary repeats) and $\widetilde{\theta}(v)$ (no repeats). We write $\overline{\mu}(v)$ for the permutation with $\mathfrak{c}(\overline{\mu}(v)) = \overline{\theta}(v)$ and set $|\overline{\theta}(v)| := \sum_i \overline{\theta}_i(v) = \ell(\overline{\mu}(v))$. The number of nonzero parts of $\overline{\theta}(v)$ equals the number of nonzero parts of $\mathfrak{c}(v)$.

Definition 4.2 ($\overset{i}{\Rightarrow}$, $\sigma_i(u,v)$, $D_i(u,v)$)**.** Let $u, v \in S_\infty$ and $i \geq 1$ be an integer. We define $u \overset{i}{\Rightarrow} v$ if there exist distinct integers $b_1, b_2, \ldots, b_k$ such that $b_j \neq i$ for all j, $\ell(ut_{i,b_1}\cdots t_{i,b_j}) = \ell(u) + j$ for all j, and

$$v = ut_{i,b_1}\cdots t_{i,b_k}$$

We define

$$\sigma_i(u,v) = \#\{j \mid b_j < i\}$$

and we define

$$D_i(u,v) = \{u(i)\} \cup \{u(j) \mid u(j) \neq v(j)\}$$

Theorem 4.3 (Formula for ordinary double Schubert polynomials)**.** *Suppose $v \in S_\infty$. Then*

$$\mathfrak{S}_v(x;z) = \sum_{(v_0,\ldots,v_k)} \prod_{i=1}^{k} (-1)^{\sigma_i(v_{i-1},v_i)} E_{\theta_i(v)-\ell(v_{i-1},v_i),\theta_i(v)}(x; z_{D_i(v_{i-1},v_i)})$$

Where $v_{i-1} \overset{i}{\Rightarrow} v_i$ for all i, $v_0 = 1$, and $v_k = v\mu(v^{-1})$.

Theorem 4.4 (Formula for quantum double Schubert polynomials)**.** *Suppose $v \in S_\infty$. Then*

$$\widetilde{\mathfrak{S}}_v(x;z) = \sum_{(v_0,\ldots,v_k)} \prod_{i=1}^{k} (-1)^{\sigma_i(v_{i-1},v_i)} E^q_{\widetilde{\theta}_i(v)-\ell(v_{i-1},v_i),\widetilde{\theta}_i(v)}(x; z_{D_i(v_{i-1},v_i)})$$

Where $v_{i-1} \overset{i}{\Rightarrow} v_i$ for all i, $v_0 = 1$, and $v_k = v\widetilde{\mu}(v^{-1})$.

Coupled with the Pieri formulas, these formulas give an algorithm for computing products of (quantum) double Schubert polynomials. We describe its classical form by building it from two ingredients: the elementary Pieri step that generates the u-side covers, and the signed path expansion on the v-side.

Definition 4.5 (k-strip cover, admissible k-chain, the (p,k)-Pieri problem)**.** Let $w \in S_n$, let $1 \le k < n$, and let $\beta \in \{1,\dots,n\} \cup \{\infty\}$. A *$k$-strip cover of* (w,β) is a transposition t_{ab} of positions a,b with $1 \le a \le k < b \le n$ such that

(1) $w \lessdot wt_{ab}$ in Bruhat order; that is, $w(a) < w(b)$ and no c with $a < c < b$ satisfies $w(a) < w(c) < w(b)$; and
(2) $w(b) < \beta$.

Its *output* is the pair $(wt_{ab},\, w(b))$. An *admissible k-chain of length p* from u is a sequence $u = w_0, w_1, \dots, w_p$ carrying bounds $\infty = \beta_0 > \beta_1 > \cdots > \beta_p$ in which (w_t, β_t) is the output of a k-strip cover of (w_{t-1}, β_{t-1}) for every t. The *(p,k)-Pieri problem* for u asks to enumerate every admissible k-chain from u of length at most p. These chains index the permutations occurring in $e_p(x_1,\dots,x_k)\,\mathfrak{S}_u$, where e_p is the elementary symmetric polynomial.

Method 4.6 (**pieri**(u,p,k))**.** Given $u \in S_n$, $1 \le k < n$, and $0 \le p \le k$, the algorithm returns a list T of pairs (w,t), one for each admissible k-chain from u of length $t \le p$, recording its endpoint w.

(1) Initialize $T \leftarrow [(u,0)]$ and a frontier $F \leftarrow [(u,\infty)]$.
(2) For $t = 1,\dots,p$, form a new frontier $F' \leftarrow [\,]$ and, for each $(w,\beta) \in F$:
   (a) compute $A \leftarrow \{\, a \in \{1,\dots,k\} \mid w(a) < \beta \,\}$;
   (b) for each $b = k+1,\dots,n$ with $w(b) < \beta$, and each $a \in A$ for which t_{ab} is a k-strip cover of (w,β), append $(wt_{ab},\, w(b))$ to F' and $(wt_{ab},\, t)$ to T.

   Then set $F \leftarrow F'$.
(3) Return T.

Lemma 4.7 (Multiplicity-freeness)**.** *For each t, distinct admissible k-chains from u of length t have distinct endpoints; equivalently, the endpoint map is a bijection from admissible k-chains of length t onto the permutations w with $w \gtrdot^k u$ of length $\ell(u)+t$ produced by the Pieri rule for $e_t(x_1,\dots,x_k)$.*

Proof. The strictly decreasing bounds $\beta_0 > \beta_1 > \cdots$ force the values moved leftward across position k to be strictly decreasing, which is exactly the condition making the e_t-Pieri expansion multiplicity-free; see [1]. Reconstructing the chain from its endpoint by peeling off these moves in decreasing order of value shows the map is injective, hence bijective onto the Pieri support. □

We now describe the v-side primitive. Its starting point is the structure of a *dominant* double Schubert polynomial. If μ is dominant with $\mathfrak{c}(\mu) = (\lambda_1,\dots,\lambda_m)$, then $\mathfrak{S}_\mu(x;z)$ factors as a product of factorial elementary symmetric polynomials of maximal degree, one per column of the shape,

$$\mathfrak{S}_\mu(x;z) = \prod_{i=1}^{m} E_{\lambda_i,\,\lambda_i}(x; z^{(i)}),$$

where $z^{(i)}$ is the shifted z-alphabet attached to column i (Definition 3.1). For a general v one writes $\mathfrak{S}_v(x;z)$ by applying divided differences in the z-variables to the dominant polynomial $\mathfrak{S}_{\mu(v^{-1})}(x;z)$; because that polynomial factors by column, the divided-difference action decouples and may be carried out one column (one $z^{(i)}$-group) at a time. Applied to a single column this produces a *signed* descending expansion. The following primitive carries it out.

Definition 4.8 (k-strip descent beneath an edge, the down-Pieri problem)**.** Let $w, \mu \in S_n$ and $1 \le k < n$. Say w' lies *beneath the edge* μ if

$$\mathrm{inv}(w'\mu) = \mathrm{inv}(\mu) - \mathrm{inv}(w'),$$

that is, $\ell(w'\mu) = \ell(\mu) - \ell(w')$. A *$k$-strip descent* of w is a Bruhat descent $w \gtrdot wt_{ab}$ realized by a transposition t_{ab} that moves the value in the pivot position k either leftward (from a position $a < k$, with sign -1) or rightward (to a position $b > k$, with sign $+1$), touching only positions still holding their original value. Given a starting label w, the edge μ, a degree p, and the column k, the *down-Pieri problem* asks for the signed list of those length-t descending k-strip chains from w ($t \le p$) whose endpoint lies beneath μ.

Method 4.9 (**kdown**(w, μ, p, k))**.** The algorithm returns a signed list D of triples (w', t, s), one for each length-t descending k-strip chain from w ($t \le p$) whose endpoint w' lies beneath μ, with $s \in \{+1, -1\}$ its sign.

(1) Initialize $D \leftarrow [\,]$; if w is already beneath μ, append $(w, 0, +1)$. Set a frontier $F \leftarrow [(w, +1)]$.
(2) For $t = 1, \dots, p$, form $F' \leftarrow [\,]$ and, for each $(w'', s) \in F$:
   (a) (leftward) for each position $a < k$ with $w''(a) = w(a)$ such that $w'' \gtrdot w''t_{a,k}$ is a Bruhat descent, set $w''' \leftarrow w''t_{a,k}$ and append $(w''', -s)$ to F'; if w''' is beneath μ, append $(w''', t, -s)$ to D;
   (b) (rightward) for each position $b > k$ with $w''(b) = w(b)$ such that $w'' \gtrdot w''t_{k,b}$ is a Bruhat descent, set $w''' \leftarrow w''t_{k,b}$ and append $(w''', +s)$ to F'; if w''' is beneath μ, append $(w''', t, +s)$ to D.

   Then set $F \leftarrow F'$.
(3) Return D.

Applied without the edge μ, the signed descending expansion produces many extraneous terms that ultimately cancel. The edge is exactly the bound that discards them: only endpoints beneath μ are retained in D. In the assembled v-side (Method 4.10 below), only these retained labels seed the next column, so each stage passes forward only labels compatible with the remaining shape.

Method 4.10 (**vpaths**(v), the v-side path dictionaries)**.** Let $\theta := \theta(v^{-1})$ with nonzero parts in columns $1, \dots, L$, and let $vmu := v\,\mu(v^{-1})$. Processing columns from $i = L$ down to $i = 1$, maintain for each column a set of live labels (initialized at column L to the single label vmu). At column i, with $k = i$ and edge $\mu_i := \mu$ of the shape truncated below column i, call **kdown**(w, μ_i, θ_i, k) for each live label w; record its signed output triples as the transitions out of w, and let their endpoints be the live labels seeding column $i - 1$. The collected transitions form the path dictionaries used by **schubmult**.

Lemma 4.11 (Correctness of the edge filter)**.** *A permutation w' produced by a descending k-strip chain from w is retained by* **kdown**(w, μ, p, k) *if and only if w' lies beneath μ, i.e.* $\mathrm{inv}(w'\mu) = \mathrm{inv}(\mu) - \mathrm{inv}(w')$*. In the assembly of Method 4.10 the retained signed labels are exactly the terms of $\mathfrak{S}_v(x; z)$ contributed through column i, so no extraneous (cancelling) label propagates to column $i - 1$.*

Proof. The retention test is the identity $\mathrm{inv}(w'\mu) = \mathrm{inv}(\mu) - \mathrm{inv}(w')$ checked in step (2); this is precisely the condition that $w' \le \mu$ with $\ell(w'\mu) = \ell(\mu) - \ell(w')$. The factorization of the dominant polynomial by column, together with the column-wise decoupling of the z-divided differences, identifies the beneath-μ endpoints at column i with the surviving monomials of $\mathfrak{S}_v(x; z)$ after processing that column; the extraneous chains are exactly those leaving the shape, which the edge discards before they seed column $i - 1$. □

Method 4.12 (The algorithm **schubmult**)**.** Fix n and let $u, v \in S_n$. The algorithm **schubmult**(u, v) computes the expansion of $\mathfrak{S}_u(x; z)\ \mathfrak{S}_v(x; z)$ in the Schubert basis as follows. Set $\theta := \theta(v^{-1})$ and let

$$L := \#\{i \mid \theta_i \neq 0\}$$

be the number of nonzero parts. The algorithm proceeds in L layers. It maintains a *front*, a collection of nodes each consisting of an intermediate permutation on the u-side paired with a label on the v-side, initialized to the single node $(u,\ e)$. At layer i it replaces each node by the nodes obtained by combining one u-side cover, generated by **pieri**$(\,\cdot\,, \theta_i, \theta_i)$ (Method 4.6), with one v-side transition, generated by **vpaths**(v) (Method 4.10), accumulating coefficients in the coefficient ring. After all L layers the surviving coefficients at v-label $v\mu(v^{-1})$ give the structure constants.

Quantum version. The same path organization gives the quantum multiplication algorithm. Its layers are indexed by the medium theta $\overline{\theta}(v^{-1})$; equal consecutive parts that are retained in this shape are handled together by the paired quantum Pieri step. On the u-side, classical covers are replaced by the quantum covers of Remark 2.1, including length-decreasing moves, whose contributions carry the corresponding q-monomials. The v-side continues to use the descending path dictionaries of Method 4.10. Thus the two algorithms share the same layer-by-layer structure, while their Pieri transitions reflect the classical or quantum multiplication rule, respectively.

The algorithmic construction is one way to compute products. We now turn to a complementary operator description, which packages the quantum multiplication rules in the Fomin–Gelfand–Postnikov difference operators.

4.1. **Difference operators and multiplication coefficients.**

Definition 4.13 ($\chi_{j;n}$, $\chi^{(n)}$)**.** Recall the definition of the commuting difference operators $\chi_{j;n}$, with $j \leq n$, defined by Fomin, Gelfand, and Postnikov, as follows:

$$\chi_{j,n} = x_j - \sum_{1 \leq i < j} q_{ij} \partial^{t_{ij}} + \sum_{j < k \leq n} q_{jk} \partial^{t_{jk}}$$

We denote the sequence $(\chi_{1;n}, \ldots, \chi_{n;n})$ with n fixed by $\chi^{(n)}$. Recall (CITATION) that if $\mathfrak{S}_v(x)$ has at most n variables, then

$$\widetilde{\mathfrak{S}}_v(\chi^{(n)})(1) = \mathfrak{S}_v(x)$$

and hence

$$\widetilde{\mathfrak{S}}_v(\chi^{(n)}; z)(1) = \mathfrak{S}_v(x; z)$$

We specify a more precise formula, as follows.

Theorem 4.14. *Suppose $v \in S_n$. Then*

$$\widetilde{\mathfrak{S}}_v(\chi^{(n)}; z) = \sum_{u \in S_n, d} q^d c^{1,d}_{u,v}(x; z) \partial^u$$

Proof. Note that if $\mathfrak{S}_u(x)$ has at most n variables, then

$$\widetilde{\mathfrak{S}}_u(\chi^{(n)}; y)\, \widetilde{\mathfrak{S}}_v(\chi^{(n)}; z) = \sum_{w,d} q^d c^{w,d}_{u,v}(y; z)\, \widetilde{\mathfrak{S}}_w(\chi^{(n)}; y)$$

We thus have that

$$(\widetilde{\mathfrak{S}}_u(\chi^{(n)}; y)\, \widetilde{\mathfrak{S}}_v(\chi^{(n)}; z))(1) = \sum_{w,d} q^d c^{w,d}_{u,v}(y; z)\, \mathfrak{S}_w(x; y)$$

This is also equal to

$$\widetilde{\mathfrak{S}}_v(\chi^{(n)}; z)(\mathfrak{S}_u(x; y))$$

due to the fact that $(\widetilde{\mathfrak{S}}_u(\chi^{(n)}; y))(1) = \mathfrak{S}_u(x; y)$. To obtain the coefficient of ∂^u in $\widetilde{\mathfrak{S}}_v(\chi^{(n)}; z)$, we set $y = x$ after applying this operator. Since $\mathfrak{S}_w(x; x) = 0$ unless $w = 1$, we have that

$$\widetilde{\mathfrak{S}}_v(\chi^{(n)}; z)(\mathfrak{S}_u(x; y))|_{y=x} = \sum_d q^d c^{1,d}_{u,v}(x; z)$$

and the result follows. □

Theorem 4.15. *Let k be an integer and let $v, w \in S_\infty$. Then*

$$\sum_d q^d c^{w,d}_{s_k,v}(y; z) = c^w_{s_k,v}(y; z) + \sum_{i \leq k < j} q_{ij}\, \mathfrak{S}_{v t_{ij} w^{-1}}(y; z)$$

Proof. We have that the desired value is equal to the coefficient of $\mathfrak{S}_w(x; y)$ in

$$\widetilde{\mathfrak{S}}_{s_k}(\chi^{(n)}; y)(\mathfrak{S}_v(x; z))$$

We have that

$$\widetilde{\mathfrak{S}}_{s_k}(\chi^{(n)}; y) = \mathfrak{S}_{s_k}(x; y) + \sum_{i \leq k < j \leq n} q_{ij} \partial^{t_{ij}}$$

Applying this operator, we obtain

$$\sum_{i \leq k < j \leq n} q_{ij}\, \mathfrak{S}_{v t_{ij}}(x; z) + \sum_w c^w_{s_k,v}(y; z)\, \mathfrak{S}_w(x; y)$$

Applying the Cauchy formula, this is equal to

$$\sum_{i \leq k < j, u} q_{ij}\, \mathfrak{S}_{v t_{ij} u^{-1}}(y; z)\, \mathfrak{S}_u(x; y) + \sum_w c^w_{s_k,v}(y; z)\, \mathfrak{S}_w(x; y)$$

Setting $u = w$ and allowing n to be sufficiently large, we obtain the result. □

5. Quantum to classical reduction for Grassmannian products

Theorem 5.1 ([3, Theorem 2.7]). *We have the following.*

(1) $c_{u,v}^{w,d}(y;y) = 0$ *unless* $\mathrm{sgn}_i(w) + \langle d \rangle_i \leq \mathrm{sgn}_i(u) + \mathrm{sgn}_i(v)$ *for all* i.
(2) *If* $\mathrm{sgn}_k(w) + \langle d \rangle_k = \mathrm{sgn}_k(u) + \mathrm{sgn}_k(v) = 2$ *for some* k, *then*

$$c_{u,v}^{w,d}(y;y) = c_{us_k,vs_k}^{ws_k,d-\delta_k}(y;y) = \begin{cases} c_{us_k,v}^{ws_k,d-\delta_k}(y;y) & \text{if } \mathrm{sgn}_k(w) = 0 \\ c_{us_k,v}^{ws_k,d}(y;y) & \text{if } \mathrm{sgn}_k(w) = 1 \end{cases}$$

This is symmetric in u *and* v.

Corollary 5.2 ([3, Corollary 2.8]). *Let* $u, v, w \in S_\infty$ *and* $d \in Q^\vee$. *Fix an index* i.

(1) *If* $\langle d \rangle_i = \mathrm{sgn}_i(u) = 0$ *and* $\mathrm{sgn}_i(w) = \mathrm{sgn}_i(v) = 1$, *then*

$$c_{u,v}^{w,d}(y;y) = c_{u,vs_i}^{ws_i,d}(y;y).$$

(2) *If* $\langle d \rangle_i = \mathrm{sgn}_i(u) = 1$ *and* $\mathrm{sgn}_i(w) = \mathrm{sgn}_i(v) = 0$, *then*

$$c_{u,v}^{w,d}(y;y) = c_{us_i,v}^{ws_i,d-\delta_i}(y;y).$$

This is symmetric in u *and* v.

Lemma 5.3. *We have that*

$$c_{u,v}^{w,d}(y;z) = \sum_{v_1 \cdot v_2 = v} c_{u,v_2}^{w,d}(y;y)\, \mathfrak{S}_{v_1}(y;z)$$

Theorem 5.4 (Quantum reduction holds for triple quantum Schubert coefficients). *We have the following.*

(1) $c_{u,v}^{w,d}(y;z) = 0$ *unless* $\mathrm{sgn}_i(w) + \langle d \rangle_i \leq \mathrm{sgn}_i(u) + \mathrm{sgn}_i(v)$ *for all* i.
(2) *If* $\mathrm{sgn}_k(w) + \langle d \rangle_k = \mathrm{sgn}_k(u) + \mathrm{sgn}_k(v) = 2$ *for some* k, *then*

$$c_{u,v}^{w,d}(y;z) = c_{us_k,vs_k}^{ws_k,d-\delta_k}(y;z) = \begin{cases} c_{us_k,v}^{ws_k,d-\delta_k}(y;z) & \text{if } \mathrm{sgn}_k(w) = 0 \\ c_{us_k,v}^{ws_k,d}(y;z) & \text{if } \mathrm{sgn}_k(w) = 1 \end{cases}$$

This is symmetric in u *and* v.

Proof. For (1), we note that since $\mathrm{sgn}_i(v_2) \leq \mathrm{sgn}_i(v)$ for any factorization $v_1 \cdot v_2 = v$, we have that $\mathrm{sgn}_i(w) + \langle d \rangle_i \leq \mathrm{sgn}_i(u) + \mathrm{sgn}_i(v_2)$ is a necessary condition for $c_{u,v_2}^{w,d}(y;y) \neq 0$. The result then follows from Lemma 5.3.

For (2), we note that if $\mathrm{sgn}_k(w) + \langle d \rangle_k = \mathrm{sgn}_k(u) + \mathrm{sgn}_k(v) = 2$, then since $\mathrm{sgn}_k(v_2) = \mathrm{sgn}_k(v)$ for any factorization $v_1 \cdot v_2 = v$ such that $c_{u,v_2}^{w,d}(y;y) \neq 0$, Theorem 5.1(2) holds for each term in the sum in Lemma 5.3, and the result follows. □

Corollary 5.5 (z-version of Corollary 5.2(2)). *Let* $u, v, w \in S_\infty$ *and* $d \in Q^\vee$, *and fix an index* i *with* $\mathrm{sgn}_i(v) = 0$. *If* $\langle d \rangle_i = \mathrm{sgn}_i(u) = 1$ *and* $\mathrm{sgn}_i(w) = 0$, *then*

$$c_{u,v}^{w,d}(y;z) = c_{us_i,v}^{ws_i,d-\delta_i}(y;z).$$

Proof. Since $\mathrm{sgn}_i(v_2) \leq \mathrm{sgn}_i(v) = 0$ for every factorization $v_1 \cdot v_2 = v$, the hypotheses of Corollary 5.2(2) hold for each term $c_{u,v_2}^{w,d}(y;y)$ appearing in Lemma 5.3, so $c_{u,v_2}^{w,d}(y;y) = c_{us_i,v_2}^{ws_i,d-\delta_i}(y;y)$. Resumming via Lemma 5.3 gives the claim. □

Lemma 5.6 (Reduction to a coroot supported at k). *Let* $u, v, w \in S_\infty$ *and let* $d \in Q^\vee$. *Suppose* v *has precisely one descent, which is* k. *If* $c_{u,v}^{w,d}(y;z) \neq 0$, *then there exist* $u', w' \in S_\infty$ *and* $d' \in Q^\vee$ *with* $0 \leq d' \leq d$ *such that*

$$c_{u,v}^{w,d}(y;z) = c_{u',v}^{w',d'}(y;z)$$

and $\langle d' \rangle_i \leq 0$ *for all* $i \neq k$. *Moreover, if* $d' \neq 0$ *then* $\langle d' \rangle_k > 0$.

Proof. Suppose $\langle d \rangle_i > 0$ for some $i \neq k$, and choose such an i to be maximal. Because v is k-Grassmannian, $\mathrm{sgn}_i(v) = 0$, so Theorem 5.4(1) forces

$$\mathrm{sgn}_i(w) + \langle d \rangle_i \leq \mathrm{sgn}_i(u) + \mathrm{sgn}_i(v) = \mathrm{sgn}_i(u) \leq 1.$$

Since $\langle d\rangle_i \geq 1$, we conclude $\langle d\rangle_i = 1$, $\mathrm{sgn}_i(u) = 1$, and $\mathrm{sgn}_i(w) = 0$. These are exactly the hypotheses of Corollary 5.5 (note $\mathrm{sgn}_i(v) = 0$), so

$$c_{u,v}^{w,d}(y;z) = c_{us_i,v}^{ws_i,d-\delta_i}(y;z).$$

Moreover $\langle d\rangle_i = 2d_i - d_{i-1} - d_{i+1} = 1$ with $d_{i\pm1} \geq 0$ forces $d_i \geq 1$, so $d - \delta_i \in Q^\vee$ with $0 \leq d - \delta_i \leq d$ and $|d-\delta_i| < |d|$. Replacing (u, w, d) by $(us_i, v \text{ fixed}, ws_i, d-\delta_i)$ and iterating, the quantity $|d|$ strictly decreases at each step, so the process terminates in a triple (u', w', d') with $\langle d'\rangle_i \leq 0$ for all $i \neq k$.

Finally, for any finitely supported $d' \in Q^\vee$ the values $\langle d'\rangle_i = 2d'_i - d'_{i-1} - d'_{i+1}$ telescope to $\sum_i \langle d'\rangle_i = 0$. Hence $\langle d'\rangle_k = -\sum_{i\neq k}\langle d'\rangle_i \geq 0$, with equality if and only if $\langle d'\rangle_i = 0$ for all i, i.e. $d' = 0$. Thus $d' \neq 0$ implies $\langle d'\rangle_k > 0$. □

6. Proof of the Pieri formula

Lemma 6.1. *Let $u \in S_\infty$ and let $i, j \geq 1$. Then*

$$(x_i - z_j)\,\widetilde{\mathfrak{S}}_u(x;y) = (y_{u(i)} - z_j)\,\widetilde{\mathfrak{S}}_u(x;y) + \sum_{\substack{p\neq i \\ \ell(ut_{ip})=\ell(u)+1}} \mathrm{sign}(p-i)\,\widetilde{\mathfrak{S}}_{ut_{ip}}(x;y)$$

$$+ \sum_{\substack{p\neq i \\ \ell(ut_{ip})=\ell(u)-2|p-i|+1}} \mathrm{sign}(p-i) q_{ip}\,\widetilde{\mathfrak{S}}_{ut_{ip}}(x;y)$$

Definition 6.2 (Descents of a cycle, $\mathrm{desc}_u(c)$)**.** Let c be a cycle and suppose $u \in S_\infty$. Write c as

$$(a_1, \ldots, a_n)$$

where $a_n > a_i$ for all $i < n$ (this is a unique way to express the cycle since the a_j are distinct). Then for any i with $1 \leq i < n$ we say that a_i is a *descent* of c with respect to u if $u(a_i) > u(a_{i+1})$. We denote the set of all descents of c with respect to u by $\mathrm{desc}_u(c)$.

Lemma 6.3. *Suppose $u, w \in S_\infty$ and let $k > 0$ be an integer. Then $u \overset{k}{\rightsquigarrow} w$ if and only if there exist pairwise disjoint cyclic permutations $c_1, \ldots, c_n$ of respective cycle lengths $p_1 + 1, \ldots, p_n + 1$ such that*

$$w = uc_1 \cdots c_n$$

where for each i there exist pairwise distinct positive integers $a_{i,1}, \ldots, a_{i,p_i}$ and b_i such that we may write the cycle c_i as

$$c_i = (a_{i,p_i}, a_{i,p_i-1}, \ldots, a_{i,1}, b_i)$$

with $a_{i,j} \leq k$ for all $1 \leq j \leq p_i$, $b_i > k$, and $b_i > b_j$ if $i < j$, and such that for each m we have

$$\ell(u, uc_1 \cdots c_m) = \sum_{i=1}^{m} \left(p_i - 2 \sum_{a \in \mathrm{desc}_u(c_i)} (b_i - a) \right)$$

and

$$\ell(u,w) + \deg q_k(u,w) = p_1 + \cdots + p_n$$

Proof. Suppose first that $u \overset{k}{\rightsquigarrow} w$. By Remark 2.1 there are transpositions $t_{a_1b_1}, \ldots, t_{a_pb_p}$ with $a_i \leq k < b_i$, the a_i pairwise distinct, and $b_1 \geq \cdots \geq b_p$, such that

$$u = u_0 \lessdot_k^{\mathfrak{q}} u_1 \lessdot_k^{\mathfrak{q}} \cdots \lessdot_k^{\mathfrak{q}} u_p = w, \qquad u_i = u_{i-1}t_{a_ib_i},$$

is a saturated chain in the quantum k-Bruhat order. Grouping the transpositions that share a common top b assembles them, exactly as in the non-quantum case [7, Lemma 7.3], into a cycle

$$c = (a_{i+r}, \ldots, a_i, b) = t_{a_ib} \cdots t_{a_{i+r}b};$$

ranging over the distinct values of b in decreasing order yields pairwise disjoint cycles $c_1, \ldots, c_n$ with $w = uc_1 \cdots c_n$ of the asserted form. To compute the length, set $c^{(j)} = t_{a_ib} \cdots t_{a_jb} = (a_j, \ldots, a_i, b)$; the step from $uc^{(j-1)}$ to $uc^{(j)} = uc^{(j-1)}t_{a_jb}$ exchanges the value $u(a_{j-1})$ in position b with the value $u(a_j)$ in position a_j. By the cover characterization of Remark 2.1 this step is a classical cover, of length increment $+1$, when

$u(a_j) < u(a_{j-1})$, and a quantum cover, of length increment $-(2(b-a_j)-1)$, when $u(a_j) > u(a_{j-1})$, that is, precisely when $a_j \in \mathrm{desc}_u(c)$. Summing these increments over the steps constituting $c_1, \ldots, c_m$ gives

$$\ell(u, uc_1 \cdots c_m) = \sum_{i=1}^{m} \left(p_i - 2 \sum_{a \in \mathrm{desc}_u(c_i)} (b_i - a) \right),$$

and, recording each quantum step in the q-degree as in Definition 3.1, $\ell(u,w) + \deg q_k(u,w) = p_1 + \cdots + p_n$.

Conversely, suppose pairwise disjoint cycles $c_1, \ldots, c_n$ as in the statement are given, and write $c_i = t_{a_{i,1}b_i} \cdots t_{a_{i,p_i}b_i}$. Applying these transpositions in order, the step from $uc^{(j-1)}$ to $uc^{(j)} = uc^{(j-1)}t_{a_{i,j}b_i}$ exchanges the values $u(a_{i,j})$ and $u(a_{i,j-1})$ in positions $a_{i,j} \le k < b_i$. Any transposition t_{ab} changes the length by at least $+1$ when it places the larger of the two exchanged values in the smaller position a, and by at least $-(2(b-a)-1)$ in every case, with equality in either bound if and only if the step is a classical, respectively quantum, cover in the sense of Remark 2.1. Applying the first bound at the non-descent steps and the second at the descents therefore gives

$$\ell(u, uc_1 \cdots c_n) \ge p_1 + \cdots + p_n - 2 \sum_{i=1}^{n} \sum_{a \in \mathrm{desc}_u(c_i)} (b_i - a),$$

and the length hypothesis of the statement forces equality throughout, so every step is a cover of the quantum k-Bruhat order and hence $u \overset{k}{\rightsquigarrow} w$. □

Corollary 6.4 (Cycle formula for the quantum weight)**.** *Suppose $u \overset{k}{\rightsquigarrow} w$, and write the nontrivial cycles in the decomposition of $u^{-1}w$ as $c_i = (a_{i,p_i}, \ldots, a_{i,1}, b_i)$, as in Lemma 6.3. Then*

$$q_k(u,w) = \prod_i \prod_{a \in \mathrm{desc}_u(c_i)} q_{ab_i}.$$

Equivalently, the exponent of q_r in $q_k(u,w)$ is

$$\#\{(i,a) : a \in \mathrm{desc}_u(c_i),\ a \le r < b_i\}.$$

Proof. In the factorization of c_i into the transpositions $t_{a_{i,1}b_i} \cdots t_{a_{i,p_i}b_i}$, the step indexed by $a_{i,j}$ is a quantum cover precisely when $u(a_{i,j}) > u(a_{i,j-1})$, that is, precisely when $a_{i,j} \in \mathrm{desc}_u(c_i)$. The recursive definition of $q_k(u,w)$ assigns the factor $q_{a_{i,j}b_i}$ to exactly that step and assigns no factor to a classical cover. Multiplying these step weights gives the first formula. Since $q_{ab} = q_a \cdots q_{b-1}$, the exponent of q_r is the number of such descents whose interval contains r, giving the second formula. □

Definition 6.5 ($\lhd$)**.** If $u, u' \in S_\infty$, we define $u \lhd u'$ if $\ell(u,u') = 1$ or $u' = ut_{ab}$ where $\ell(u,u') = 1 - 2(b-a)$. Equivalently, $u \lhd ut_{ab}$ exactly when $u \lessdot_k^{\mathbf{q}} \mathbf{q}^\gamma ut_{ab}$ is a cover of the quantum k-Bruhat order for some (equivalently, any) k with $a \le k < b$, in the sense of Remark 2.1.

Lemma 6.6. *Suppose $u, w \in S_\infty$ and let $k > 1$ be an integer. Suppose $u \overset{k}{\rightsquigarrow} w$ and let $c = (a_p, \ldots, a_1, b)$ be a cycle as in Lemma 6.3 in the decomposition of $u^{-1}w$. Then* $\mathrm{desc}_u(c)$ *has at most one element.*

Proof. This is clearly true if $p = 1$, or if the result holds for $p-1$ but $c^{(p-1)}$ has no descents. Suppose then that $c^{(p-1)}$ has a descent, say a_i for exactly one $i < p$ by the induction hypothesis, and that $u(a_p) > u(a_{p-1})$, meaning that $a_p \in \mathrm{desc}_u(c^{(p-1)})$. a_i being a descent would require that

$$u(a_i) > uc^{(i-1)}(j) > u(a_{i-1})$$

for all j with $a_i < j < b$. Furthermore, we also have that for all j with $a_p < j < b$ that

$$u(a_p) > uc^{(p-1)}(j) > u(a_{p-1})$$

Suppose $a_p < a_i$. Then $u(a_p) > uc^{(p-1)}(a_i) = u(a_{i-1}) > u(a_{p-1})$. This means that necessarily $a_{p-1} < a_i$, since $u(a_{i-1}) = uc^{(i)}(a_i)$ is minimal in the range $a_i \le j \le b$ in $uc^{(i)}$. For the same reason, we cannot have that $uc^{(p-2)}(b) < u(a_{i-1})$, because this would have to come from some $t_{a_q b}$ with $i < q < p-1$, $a_q < a_i$, and and $u(a_q) < u(a_{i-1})$, which would increase the length by more than 1. However, since $uc^{(p-2)}(b) > u(a_{i-1}) > u(a_{p-1})$, this would imply that $t_{a_{p-1}b}$ increases the length by more than 1, which is a contradiction. Thus we cannot have a descent a_p with $a_p < a_i$.

If instead $a_p > a_i$, then $u(a_i) > u(a_p) > u(a_{i-1})$. Since a_p is a descent, $uc^{(p-1)}(b) = u(a_{p-1}) < u(a_p)$, but $uc^{(i)}(b) = u(a_i) > u(a_p)$. This would imply that there is some q with $i < q \leq p-1$ such that $a_q > a_p$ and $uc^{(q-1)}t_{a_q b}(b) = u(a_q) < u(a_p)$, because if we had $a_q < a_p$ then the length would increase by more than 1. Let q be minimal with this property. In that case, $u(a_{q-1}) > u(a_p)$ and $uc^{(p-1)}(a_q) = u(a_{q-1})$, contradicting the fact that a_p is a descent.

Having excluded the cases $a_p > a_i$ and $a_p < a_i$, by the assumption that $a_p \neq a_i$ it follows by induction that there can be at most one descent. □

Definition 6.7 (Redefinition of $\mathrm{desc}_u(c)$)**.** If c is a cycle, we redefine $\mathrm{desc}_u(c)$ to be the unique element of the previously defined set $\mathrm{desc}_u(c)$ if it exists, or otherwise 0.

Lemma 6.8. *Suppose $u, w \in S_\infty$ and let $k > 1$ be an integer. Suppose $u \overset{k}{\rightsquigarrow} w$ and let $c = (a_p, \ldots, a_1, b)$ be a cycle as in Lemma 6.3 in the decomposition of $u^{-1}w$, with $a_q = k$ being a descent. Then for all $j \geq q$ we have that $u(a_j) > u(b)$.*

Proof. We have that $u(a_q) > u(a_{q-1})$. By Lemma 6.6, k is the only descent of c. We claim that for all r with $k \leq r < b$ we have that $u(r) > u(b)$. This is because for all such r we have that $u(a_{q-1}) < u(r)$ since a_q is a descent, and if $u(b) > u(r)$ for some such r then we may pick a maximal $\ell < q$ such that $u(a_\ell) > u(r)$ for some such r. Then $u(a_{\ell+1}) < u(r)$ for all such r, and the reflection $t_{a_{\ell-1}b}$ would increase the length by more than 1, which cannot happen. For the same reason, we cannot have that $u(a_m) < u(r)$ for any r with $k < r \leq b$, since $u(a_q) > u(r)$ for all such r and hence the reflection $t_{a_{m-1}a_m}$ would increase the length by more than q. Thus since in particular $u(r) > u(b)$ for all such r, we must have that $u(a_j) > u(b)$ for all $j \geq q$, as desired. □

Lemma 6.9. *Suppose $u, w \in S_\infty$ and let $k > 1$ be an integer. If $u \overset{k}{\rightsquigarrow} w$ but $u \overset{k-1}{\not\rightsquigarrow} w$, then there is a unique $b > k$ such that $wt_{kb} \lhd w$ and $u \overset{k-1}{\rightsquigarrow} wt_{kb}$. Furthermore, in that case there does not exist a $a < k$ such that $wt_{ak} \lhd w$ and $u \overset{k-1}{\rightsquigarrow} wt_{ak}$.*

Proof. Suppose $u \overset{k}{\rightsquigarrow} w$ but $u \overset{k-1}{\not\rightsquigarrow} w$. Let $c_1, \ldots, c_n$ be the pairwise disjoint cycles in the decomposition of $u^{-1}w$, with the cycle length of c_i being $p_i + 1$. Some such cycle must contain k, as otherwise we would have $u \overset{k-1}{\rightsquigarrow} w$ by Lemma 6.3. Assume that $c_i = (a_{i,p_1}, \ldots, a_{i,1}, b_i)$ is the cycle that contains k and set $b = b_i$. We claim that $u \overset{k-1}{\rightsquigarrow} wt_{kb}$ and $wt_{kb} \lhd w$. Note that $w = (wt_{kb})t_{kb}$ and hence

$$(uc_1 \cdots c_n t_{kb})t_{kb} = w$$

Let j be the index such that $a_j = k$. Then we have

$$\begin{aligned}(a_{i,p_i}, \ldots, a_{i,j+1}, k, a_{i,j-1}, \ldots, a_{i,1}, b_i)t_{kb} &= (a_{i,j-1}, \ldots, a_{i,1}, b, a_{i,p_i}, \ldots, a_{i,j+1}, k)t_{kb} \\ &= (a_{i,j-1}, \ldots, a_{i,1}, b)(a_{i,p_i}, \ldots, a_{i,j+1}, k)\end{aligned}$$

Note that $wt_{kb}(k) = w(b) = u(a_{i,p_i})$ and $wt_{kb}(b) = w(k) = u(a_{i,j-1})$. We need that

$$\ell(u, wt_{kb}) = p_i' + p_i'' - 2(b - \mathrm{desc}_u(c_i')) - 2(k - \mathrm{desc}_u(c_i'')) + \ell(u, uc_1 \cdots c_{i-1}c_{i+1} \cdots c_m)$$

We have that

$$\ell(u, w) = p_i - 2(b - \mathrm{desc}_u(c_i)) + \ell(u, uc_1 \cdots c_{i-1}c_{i+1} \cdots c_m)$$

and

$$\ell(u, wt_{kb}) \geq p_i' + p_i'' - 2(b - \mathrm{desc}_u(c_i')) - 2(k - \mathrm{desc}_u(c_i'')) + \ell(u, uc_1 \cdots c_{i-1}c_{i+1} \cdots c_m)$$

If c_i' has a descent, then c_i'' does not, and we are done. If c_i'' has a descent, then

$$w(k) = u(a_{i,j-1}) < u(b) < u(a_{i,p_i}) = w(b)$$

by Lemma 6.8, so

$$\ell(w) < \ell(wt_{bk}) \leq \ell(w) + 2(b - k) - 1$$

so

$$\ell(u, wt_{kb}) = \ell(wt_{bk}) - \ell(u) \leq p_i' + p_i'' - 2(b - \mathrm{desc}_u(c_i')) - 2(b - \mathrm{desc}_u(c_i'')) + 2(b - k) + \ell(u, uc_1 \cdots c_{i-1}c_{i+1} \cdots c_m)$$

Since we already have the opposite inequality, we are done if c_i'' has a descent, since $wt_{bk} \lhd w$. If neither has a descent but c_i does, then this descent must be at k. We have that $wt_{bk}(k) = u(a_{i,p_i}) > u(b)$ by Lemma

6.8, and $wt_{bk}(b) = u(a_{i,j-1}) < u(b)$. Furthermore, $u(k) > u(b) > u(r) > u(a_{i,j-1})$ for all r with $k < r < b$ since k is a descent. Thus $wt_{bk} \lhd w$.

$a_{i,q} < r < b$, so the length decreases by $2(b - a_{i,q}) - 1$ as desired. If $u(a_{i,j+1}) > u(k)$

The uniqueness of b among the indices $r > k$ with $\ell(ut_{kr}) = \ell(u) + 1$, and the nonexistence of an index $q < k$ with $\ell(ut_{qk}) = \ell(u) + 1$ and $ut_{qk} \overset{k-1}{\rightsquigarrow} w$, now follow exactly as in the proof of [7, Lemma 7.6]. Those arguments use only the disjoint cycle characterization of the Pieri relation, which in the present setting is supplied by Lemma 6.3 in place of [7, Lemma 7.3]: in each case the reflection t_{kr} (respectively t_{qk}) either merges the cycle c_i containing k with another cycle or splits it so that k and a second index exceeding $k-1$ lie in a common cycle of $(ut_{kr})^{-1}w$ (respectively $(ut_{qk})^{-1}w$), whence $ut_{kr} \overset{k-1}{\not\rightsquigarrow} w$ (respectively $ut_{qk} \overset{k-1}{\not\rightsquigarrow} w$) by Lemma 6.3. □

We organize the inductive step of Theorem 3.2 around the position k, distinguishing whether or not $u(k) = w(k)$. The following lemma records the two combinatorial facts about the interaction of $\overset{k}{\rightsquigarrow}$ and $\overset{k-1}{\rightsquigarrow}$ at position k that make this dichotomy exhaustive.

Lemma 6.10. *Suppose $u, w \in S_\infty$ and let $k > 1$ be an integer.*

(i) *If $u(k) = w(k)$, then $u \overset{k}{\rightsquigarrow} w$ if and only if $u \overset{k-1}{\rightsquigarrow} w$.*

(ii) *If $u(k) \neq w(k)$ and $u \overset{k}{\rightsquigarrow} w$, then $u \overset{k-1}{\not\rightsquigarrow} w$.*

Proof. Both statements follow from the cycle description of Lemma 6.3, according to which $u \overset{m}{\rightsquigarrow} w$ if and only if in the disjoint cycle decomposition of $u^{-1}w$ each cycle contains exactly one index exceeding m, subject to the length conditions. If $u(k) = w(k)$ then k is a fixed point of $u^{-1}w$, hence lies in no nontrivial cycle; the constraint distinguishing level k from level $k-1$ concerns only the indices exceeding $k-1$ that occur in the cycles, and k is not among them. Thus the cycle conditions for level k and level $k-1$ coincide, proving (i). For (ii), if $u(k) \neq w(k)$ then k lies in a nontrivial cycle c of $u^{-1}w$; if $u \overset{k}{\rightsquigarrow} w$, the level-$k$ condition forces k to be the unique index of c exceeding $k-1$ that is admissible at level k, but then c contains an index $b > k$ as its cycle top, so c contains two indices $\geq k$, violating the level-$(k-1)$ condition. Hence $u \overset{k-1}{\not\rightsquigarrow} w$. □

The heart of the induction is the following cancellation, which replaces the informal "the sums are empty" step of the classical argument. When $u(k) = w(k)$ the diagonal term alone already reproduces $c_k(u, w)$, by adjoining the variable $y_{u(k)}$ to the product $\mathbf{E}(y_{\mathrm{fix}_{k-1}}; z)$; the reflection sums and the quantum term must therefore cancel among themselves.

Lemma 6.11 (Reflection–drop cancellation)**.** *Suppose $u, w \in S_\infty$, let $k > 1$, and suppose $u(k) = w(k)$. Then*

$$\sum_{a<k,\ wt_{ak}\lhd w} \operatorname{sign}(a-k)\,\kappa_{ak}\,c_{k-1}(u, wt_{ak}) \;+\; \sum_{b>k,\ wt_{kb}\lhd w} \operatorname{sign}(b-k)\,\kappa_{kb}\,c_{k-1}(u, wt_{kb}) \;+\; q_{k-1}\,c_{k-2}(u, w) = 0,$$

where $\kappa_{rs} = q_{rs}$ if $wt_{rs} \lhd w$ is length-decreasing and $\kappa_{rs} = 1$ if it is length-increasing. In fact the two reflection sums contribute at most one nonzero summand in total: it occurs precisely when $u \overset{k-2}{\rightsquigarrow} w$ (equivalently, when the quantum term is nonzero), it is indexed by some $a < k$, and it equals $-q_{k-1}c_{k-2}(u, w)$.

Proof. Throughout we use that $u(k) = w(k)$, so that k is a fixed point of $\pi := u^{-1}w$.

Step 1: no reflection t_{kb} with $b > k$ contributes. Suppose $b > k$ and $wt_{kb} \lhd w$ with $c_{k-1}(u, wt_{kb}) \neq 0$, so $u \overset{k-1}{\rightsquigarrow} v$ for $v := wt_{kb}$. Then $u^{-1}v = \pi t_{kb}$; since k is a fixed point of π and $b > k$, right multiplication by t_{kb} adjoins k to the π-cycle containing b (or creates the 2-cycle $(k\,b)$). Either way the resulting cycle of $u^{-1}v$ contains two indices exceeding $k-1$, namely k and an index $\geq b > k$, contradicting the level-$(k-1)$ condition of Lemma 6.3. Hence only reflections t_{ak} with $a < k$ can contribute, and each such summand carries the sign $\operatorname{sign}(a-k) = -1$.

Step 2: the reflection sum has at most one nonzero term, occurring precisely when $u \overset{k-2}{\rightsquigarrow} w$. A reflection t_{ak} with $a < k$ contributes precisely when $u \overset{k-1}{\rightsquigarrow} v$ for $v := wt_{ak}$; here $u^{-1}v = \pi t_{ak}$, so t_{ak} adjoins the fixed point k to the π-cycle containing a (or creates the 2-cycle $(a\,k)$ when a is fixed by π). By Lemma 6.3, $u \overset{k-1}{\rightsquigarrow} v$ requires this enlarged cycle to contain exactly one index exceeding $k-1$; as it already contains k, it may contain no further index $\geq k$. Since k is fixed by π, an index occurring in a nontrivial π-cycle and

exceeding $k-1$ is in fact $\geq k+1$, unless it equals $k-1$. Thus the enlarged cycle is admissible only if the π-cycle receiving k has all indices $\leq k-1$, i.e. its top equals $k-1$; the π-cycles being disjoint, at most one $a<k$ contributes. Such a configuration exists exactly when $u \overset{k-2}{\rightsquigarrow} w$: if $u \overset{k-2}{\rightsquigarrow} w$ and $k-1$ is not fixed by π, then the π-cycle through $k-1$ has top $k-1$ (any second index $\geq k-1$ would be $\geq k+1$, violating the level-$(k-2)$ condition), while if $k-1$ is fixed the choice $a=k-1$ creates the 2-cycle $(k-1\,k)$; conversely a contributing t_{ak} forces a π-cycle of top $k-1$ (or $k-1$ fixed with $a=k-1$), whence $u \overset{k-2}{\rightsquigarrow} w$. This is exactly the condition under which the quantum term is nonzero.

Step 3: the contributing term cancels the quantum term. Assume $u \overset{k-2}{\rightsquigarrow} w$, let t_{a^*k} ($a^*<k$) be the unique contributing reflection, and set $v := wt_{a^*k}$. By Step 2 the cycle of $u^{-1}v$ containing k has top k. Since v agrees with w off $\{a^*,k\}$ and $v(a^*)=w(k)=u(k)$, both a^* and $k-1$ leave the agreement set in passing to level $k-1$, and comparison of the invariants of Definition 3.1 gives

$$\mathrm{fix}_{k-1}(u,v)=\mathrm{fix}_{k-2}(u,w), \qquad n_{k-1}(u,v)=n_{k-2}(u,w)+1$$

(using $n_m=m-|\mathrm{fix}_m|$). Hence $c_{k-1}(u,v)$ and $c_{k-2}(u,w)$ carry the *same* factorial elementary polynomial $\mathbf{E}(y_P;z)$ with $P=\mathrm{fix}_{k-2}(u,w)$. Finally the q-weights match. Write $\kappa_{a^*k}=1$ if t_{a^*k} is length-increasing and $\kappa_{a^*k}=q_{a^*k}$ if length-decreasing. As the q-weight is multiplicative over the cycles of $u^{-1}(\cdot)$ and only the cycle receiving k is altered (from top $k-1$ to top k), the definition of the weights yields

$$\kappa_{a^*k}\, q_{k-1}(u,v)=q_{k-1}\, q_{k-2}(u,w).$$

Therefore the contributing summand is

$$\mathrm{sign}(a^*-k)\,\kappa_{a^*k}\,c_{k-1}(u,v)=-\,\kappa_{a^*k}\,q_{k-1}(u,v)\,\mathbf{E}(y_P;z)=-\,q_{k-1}\,q_{k-2}(u,w)\,\mathbf{E}(y_P;z)=-\,q_{k-1}c_{k-2}(u,w),$$

cancelling the quantum term. When $u \overset{k-2}{\not\rightsquigarrow} w$ the quantum term vanishes and, by Step 2, no reflection contributes; all three terms are 0. In every case the displayed sum is 0. □

Remark 6.12. The content of Lemma 6.11 is the identification of the quantum term $q_{k-1}c_{k-2}(u,w)$ with the unique level-$(k-1)$ neighbor v satisfying $u \overset{k-1}{\rightsquigarrow} v$, $u \overset{k-2}{\not\rightsquigarrow} v$, $\mathrm{fix}_{k-1}(u,v)=\mathrm{fix}_{k-2}(u,w)$ and $n_{k-1}(u,v)=n_{k-2}(u,w)+1$, obtained by adjoining the fixed point k to the π-cycle of top $k-1$. The connecting reflection t_{a^*k} may be length-increasing or length-decreasing; in either case its weight κ_{a^*k} satisfies the balance $\kappa_{a^*k}q_{k-1}(u,v)=q_{k-1}q_{k-2}(u,w)$. Uniqueness of v is a consequence of the cycle structure (there is a single π-cycle of top $k-1$); this intrinsic matching by the invariants (P,n) is what the $u(k)\neq w(k)$ branch of Lemma 6.14 invokes.

When $u(k)\neq w(k)$ the position k lies in a nontrivial cycle of $\pi=u^{-1}w$, and the reflection sums of the vanishing identity below no longer telescope against the diagonal; instead they cancel among themselves and against the quantum term, one q-weight at a time. The following lemma isolates this cancellation. It is the moved-case counterpart of Lemma 6.11, and its non-quantum instances ($M=1$) reduce to the pairing of an up-neighbor against a down-neighbor effected by [7, Lemma 7.9] in the cancellation of the classical Pieri rule [7, Proof of Proposition 4.4].

Lemma 6.13 (Moved reflection cancellation)**.** *Let $u,w\in S_\infty$, let $k>1$, and suppose $u \overset{k}{\not\rightsquigarrow} w$ with $u(k)\neq w(k)$. Let $\pi=u^{-1}w$, let C be the π-cycle containing k — nontrivial, since $u(k)\neq w(k)$ — and set $B=\max(C)$. Fix a q-monomial $q^M\neq q_{k-1}(u,w)$, and let $\mathcal{G}_M$ be the set of Monk summands of* (6.3)–(6.4) *of q-weight q^M, augmented by the quantum term $q_{k-1}c_{k-2}(u,w)$ of* (6.5) *when $q^M=q_{k-1}q_{k-2}(u,w)$. Then the summands of $\mathcal{G}_M$ sum to* 0.

Proof. A neighbor $v=wt$, with $t\in\{t_{ak}:a<k\}\cup\{t_{kb}:b>k\}$, contributes to $\mathcal{G}_M$ only if $u \overset{k-1}{\rightsquigarrow} v$; writing $\sigma=u^{-1}v=\pi t$, Lemma 6.3 requires every cycle of σ to contain exactly one index exceeding $k-1$. Since t moves π only at its two positions, one of which is $k\in C$, the passage $\pi\rightsquigarrow\sigma$ either *splits* C (when the other position lies in C) or *merges* C with the component of the other position (when it does not); every other π-cycle is unchanged and so must already satisfy the level-$(k-1)$ condition.

Up-neighbors are unique and force C to have exactly the indices k,B above $k-1$. Let $v=wt_{kb}$ with $b>k$. If $b\notin C$, the merge produces a cycle containing k, b and $B\geq b$, hence two indices exceeding $k-1$: inadmissible. Thus $b\in C$, and t_{kb} splits C into a piece through k and a piece through b; admissibility forces each to hold a single index above $k-1$, so b is that index for its piece. As $B=\max(C)>k-1$, this

yields $b = B$ and shows C has exactly the two indices k, B above $k-1$, with $B > k$. Hence a contributing up-neighbor, if present, is unique and equals wt_{kB}.

Down-neighbors. Let $v = wt_{ak}$ with $a < k$. If $a \in C$, right multiplication by t_{ak} splits C; for $u \overset{k-1}{\rightsquigarrow} v$ each resulting nontrivial cycle must contain exactly one index greater than $k-1$. Thus a cycle C containing three or more such indices cannot contribute. If C contains exactly k and $B > k$ among its indices greater than $k-1$, a contributing split must separate these two indices. If C contains only k among its indices greater than $k-1$ (so $B = k$), a contributing split can only separate a fixed point from the cycle. If $a \notin C$, the transposition joins C to the cycle containing a. The joined cycle must still contain exactly one index greater than $k-1$, namely k; hence this case is possible only when $B = k$ and the other cycle has all its indices at most $k-1$. If that other cycle is nontrivial, its top must be $k-1$.

(1) The case $B = k$. There are two distinct boundary operations. A split which peels off a singleton leaves the other nontrivial cycle with top k; its q-weight is $q_{k-1}(u, w)$ and it cancels the diagonal contribution (the case treated separately in Lemma 6.14(i)). A join, on the other hand, attaches to the cycle of top k either a fixed point or the unique other nontrivial cycle of top $k-1$. It is this join, when it is a Bruhat cover, that matches the quantum term. For $v = wt_{ak}$, the cycles of $u^{-1}v$ are obtained by this join. The cycle condition for $u \overset{k-2}{\rightsquigarrow} w$ is exactly the condition that the new cycle at level $k-1$ has its required single index above $k-1$; the cover condition selects at most one such a. Comparing fixed positions and applying Corollary 6.4 to the cycle factorizations gives

$$\mathrm{fix}_{k-1}(u, v) = \mathrm{fix}_{k-2}(u, w), \qquad n_{k-1}(u, v) = n_{k-2}(u, w) + 1, \qquad \kappa_{ak} q_{k-1}(u, v) = q_{k-1} q_{k-2}(u, w),$$

where the last equality is an equality of monomials: by Corollary 6.4, its q_r-exponent on either side is the number of quantum descents whose interval contains r, including the interval contributed by the length-decreasing Monk edge when present. This is precisely the change in the descent intervals when the two cycles are joined (or when the fixed point is attached). The uniqueness is the uniqueness of the admissible cover in the cycle containing k.

(2) The case $B > k$. A contributing down-neighbor must split the cycle containing k from the one containing B. The Bruhat-cover condition leaves exactly one such split: it is the cut immediately before the unique descent of this cycle (Lemma 6.6); the length-decreasing alternative is possible only for this cut, by Lemma 6.8. Thus there is at most one down-neighbor, and it is paired with the unique up-neighbor wt_{kB}.

Cancellation. Suppose first $q^M = q_{k-1} q_{k-2}(u, w)$, so the quantum term lies in $\mathcal{G}_M$. If it is nonzero, then $u \overset{k-2}{\rightsquigarrow} w$; the cycle condition forces $B = k$, and there is no up-neighbor. By (1) the unique join down-neighbor v has $\mathrm{fix}_{k-1}(u, v) = \mathrm{fix}_{k-2}(u, w)$. Thus $c_{k-1}(u, v)$ and $c_{k-2}(u, w)$ have the same factorial factor, and the down summand, whose sign is -1, is exactly $-q_{k-1} c_{k-2}(u, w)$; it cancels the quantum term. If there is no such v, the quantum term is zero as well.

Now suppose $q^M \neq q_{k-1}(u, w), q_{k-1} q_{k-2}(u, w)$. By (1) every merge or peel down has one of the two excluded weights, so every summand of $\mathcal{G}_M$ is of split type with $B > k$. In this case the up-neighbor and the unique down-neighbor have the same fixed-position set $P = \mathrm{fix}_{k-1}(u, w)$: the up-neighbor agrees with w below k, and the down-neighbor changes the value at a^* to $w(k)$, which is not $u(a^*)$ (otherwise the split would be the fixed-point peel of case (1)). Their factorial factors are therefore both $\mathbf{E}(y_P; z)$. For the quantum weights, the two splits have the same resulting cycle tops k and B; their descent intervals agree except that the interval between k and B is supplied by the Monk factor on the length-decreasing edge. Applying the interval-count formula of Corollary 6.4 gives

$$\kappa_{kB} q_{k-1}(u, wt_{kB}) = \kappa_{a^* k} q_{k-1}(u, wt_{a^* k}) = q^M.$$

Since the two terms have opposite signs, they cancel. If neither neighbor exists, the corresponding collection is empty. Hence $\mathcal{G}_M = 0$. □

The remaining case of the induction is when $u \overset{k}{\not\rightsquigarrow} w$, where the target coefficient is 0.

Lemma 6.14 (Vanishing)**.** *Suppose $u, w \in S_\infty$, let $k > 1$, and suppose $u \overset{k}{\not\rightsquigarrow} w$. Then*

$$(y_{w(k)} - z_1)\, c_{k-1}(u, w) + \sum_{b > k,\ wt_{kb} \lhd w} \kappa_{kb}\, c_{k-1}(u, wt_{kb}) - \sum_{a < k,\ wt_{ak} \lhd w} \kappa_{ak}\, c_{k-1}(u, wt_{ak}) + q_{k-1} c_{k-2}(u, w) = 0,$$

where $\kappa_{rs} = q_{rs}$ if $wt_{rs} \lhd w$ is length-decreasing and $\kappa_{rs} = 1$ if it is length-increasing.

Proof. Suppose first that $u(k) = w(k)$. Then by Lemma 6.10(i) also $u \overset{k-1}{\not\rightarrow} w$, so $c_{k-1}(u, w)$ vanishes; hence the diagonal term is 0, and the remaining reflection sums and the quantum term cancel by Lemma 6.11. This settles the case $u(k) = w(k)$ completely.

Now suppose $u(k) \neq w(k)$. Since $u \overset{k}{\not\rightarrow} w$, Lemma 6.3 shows the cycle of $u^{-1}w$ containing k has a top $b > k$, so $c_k(u, w) = 0$; writing F for the right-hand side (6.2)–(6.5), we must show $F = 0$.

We grade by the cohomological degree $\deg q_i = 2$. Under this grading (6.1) is homogeneous, and the q_i are algebraically independent over $\mathbb{Z}[y, z]$. By the definition of c_m, each summand of F is a monomial in the q_i times a factorial elementary polynomial in $\mathbb{Z}[y, z]$; collecting by q-monomial we write $F = \sum_M q^M F_M$ with $F_M \in \mathbb{Z}[y, z]$, and

$$F = 0 \iff F_M = 0 \text{ for every } q\text{-monomial } M.$$

Fix M and let $\mathcal{G}_M$ be the collection of summands of F whose q-weight equals q^M; here the diagonal (6.2) carries weight $q_{k-1}(u, w)$, the quantum term carries $q_{k-1}q_{k-2}(u, w)$, and the Monk summand of (6.3)–(6.4) reaching $w' = wt_{ak}$ or $w' = wt_{kb}$ carries $\kappa\, q_{k-1}(u, w')$ with $\kappa \in \{1, q_{ak}, q_{kb}\}$ its drop-weight. We show $F_M = 0$ in two exhaustive cases.

Case (i): $\mathcal{G}_M$ contains the diagonal. It carries weight $q_{k-1}(u, w)$, so $q^M = q_{k-1}(u, w)$; in particular $u \overset{k-1}{\rightsquigarrow} w$. Write $P = \mathrm{fix}_{k-1}(u, w)$ and $n = n_{k-1}(u, w)$. Set $a_0 = u^{-1}(w(k))$; since $u(k) \neq w(k)$ and $w(k)$ is a value of u occurring at a position $< k$ (this holds whenever the diagonal is nonzero), we have $a_0 < k$. The Monk summand indexed by $t_{a_0 k}$ then satisfies $u \overset{k-1}{\rightsquigarrow} wt_{a_0 k}$ with

$$\mathrm{fix}_{k-1}(u, wt_{a_0 k}) = P \cup \{w(k)\}, \qquad n_{k-1}(u, wt_{a_0 k}) = n - 1,$$

carries weight $q_{k-1}(u, w)$, and is the *only* Monk summand in $\mathcal{G}_M$. Dividing $\mathcal{G}_M$ by the common factor $q_{k-1}(u, w)$, its two contributions read

$$(y_{w(k)} - z_1)\, \mathbf{E}(y_P; z) - \mathbf{E}(y_{P\cup\{w(k)\}}; z),$$

which vanish since $\mathbf{E}(y_{P\cup\{w(k)\}}; z) = (y_{w(k)} - z_1)\mathbf{E}(y_P; z)$ by adjoining the variable $y_{w(k)}$. Hence $F_M = 0$.

Case (ii): $\mathcal{G}_M$ does not contain the diagonal. Then $q^M \neq q_{k-1}(u, w)$, and $\mathcal{G}_M$ consists of Monk summands of (6.3)–(6.4) together with the quantum term when $q^M = q_{k-1}q_{k-2}(u, w)$. By Lemma 6.13 these summands sum to 0, so $F_M = 0$.

As the two cases are exhaustive, every $F_M = 0$, so $F = 0$. □

We now record the recurrence satisfied by the quantum factorial elementary polynomials, which drives the induction.

Lemma 6.15. *For all integers k we have*

$$\widetilde{E}_{k,k}(x; z) = (x_k - z_1)\widetilde{E}_{k-1,k-1}(x; z) + q_{k-1}\widetilde{E}_{k-2,k-2}(x; z),$$

with the conventions $\widetilde{E}^q_{0,0} = 1$ and $\widetilde{E}_{k,k} = 0$ if $k < 0$. Moreover, the ordinary polynomials $\mathbf{E}(y_A; z) = \prod_{i\in A}(y_i - z_1)$ satisfy, for any set A and any $c \notin A$, the single-variable identity

$$\mathbf{E}(y_{A\cup\{c\}}; z) = (y_c - z_1)\, \mathbf{E}(y_A; z).$$

The first recurrence is the quantization of the classical relation $E_{k,k} = (x_k - z_1)E_{k-1,k-1}$; the extra term $q_{k-1}\widetilde{E}_{k-2,k-2}$ is the only feature absent from the non-quantum setting. Discarding all quantum terms recovers the classical Pieri formula [7, Theorem 7.1], whose term-by-term cancellation is carried out in [7, Proof of Proposition 4.4] by means of the reflection-matching Lemmas 7.6–7.9 of that paper; we adapt that argument, the essential new points being the length-decreasing steps permitted by Lemma 6.3 and the attendant bookkeeping of the powers of q.

Proof of Theorem 3.2. For a permutation w, write $[\widetilde{\mathfrak{S}}_w]F$ for the coefficient of $\widetilde{\mathfrak{S}}_w(x; y)$ in an expansion F of a quantum double Schubert polynomial in the basis $\{\widetilde{\mathfrak{S}}_w(x; y)\}$. With this notation the theorem is the assertion that, for all u and all $k \geq 1$,

$$[\widetilde{\mathfrak{S}}_w]\big(\,\widetilde{\mathfrak{S}}_u(x; y)\widetilde{E}_{k,k}(x; z)\big) = \begin{cases} q_k(u, w)\, \mathbf{E}(y_{\mathrm{fix}_k(u,w)}; z) & \text{if } u \overset{k}{\rightsquigarrow} w, \\ 0 & \text{otherwise.} \end{cases}$$

For brevity, when $u \overset{m}{\rightsquigarrow} w$ we set

$$c_m(u,w) = q_m(u,w)\,\mathbf{E}(y_{\mathrm{fix}_m(u,w)};z),$$

and $c_m(u,w)=0$ if $u \overset{m}{\not\rightsquigarrow} w$. The claim is that $[\widetilde{\mathfrak{S}}_w](\widetilde{\mathfrak{S}}_u\,\widetilde{E}_{k,k}) = c_k(u,w)$.

We argue by strong induction on k.

Base case $k=1$. We have $\widetilde{E}_{1,1}(x;z)=x_1-z_1$, and Lemma 6.1 with $i=j=1$ gives

$$(x_1-z_1)\,\widetilde{\mathfrak{S}}_u(x;y) = (y_{u(1)}-z_1)\,\widetilde{\mathfrak{S}}_u(x;y) + \sum_{\substack{b>1\\ \ell(ut_{1b})=\ell(u)+1}} \widetilde{\mathfrak{S}}_{ut_{1b}}(x;y) + \sum_{\substack{b>1\\ \ell(ut_{1b})=\ell(u)-2(b-1)+1}} q_{1b}\,\widetilde{\mathfrak{S}}_{ut_{1b}}(x;y).$$

The permutations w with $u \overset{1}{\rightsquigarrow} w$ are exactly $w=u$ and $w=ut_{1b}$ for $b>1$ with $u \lhd ut_{1b}$, since Definition 3.1 permits at most one reflection when $k=1$. For $w=u$ we have $q_1=1$, $\mathrm{fix}_1(u,u)=\{u(1)\}$, and $\mathbf{E}(y_{\{u(1)\}};z)=y_{u(1)}-z_1$, matching the diagonal term. For $w=ut_{1b}$ we have $\mathrm{fix}_1=\emptyset$ and $\mathbf{E}(y_\emptyset;z)=1$; if the step is length-increasing then $q_1(u,w)=1$, matching the second sum, and if it is length-decreasing then $q_1(u,w)=q_{1b}$, matching the third sum. All other coefficients vanish on both sides. This proves the base case.

Inductive step. Fix $k>1$ and assume the claim at all levels $m<k$. By Lemma 6.15,

$$\widetilde{\mathfrak{S}}_u\,\widetilde{E}_{k,k} = (x_k-z_1)\big(\widetilde{\mathfrak{S}}_u\,\widetilde{E}_{k-1,k-1}\big) + q_{k-1}\,\widetilde{\mathfrak{S}}_u\,\widetilde{E}_{k-2,k-2}. \tag{6.1}$$

By the inductive hypothesis, $\widetilde{\mathfrak{S}}_u\,\widetilde{E}_{k-1,k-1} = \sum_v c_{k-1}(u,v)\,\widetilde{\mathfrak{S}}_v$. Applying Lemma 6.1 with $i=k$, $j=1$ to each $\widetilde{\mathfrak{S}}_v$ and extracting the coefficient of $\widetilde{\mathfrak{S}}_w$ from (6.1), we obtain

$$[\widetilde{\mathfrak{S}}_w]\big(\widetilde{\mathfrak{S}}_u\,\widetilde{E}_{k,k}\big) = (y_{w(k)}-z_1)\,c_{k-1}(u,w) \tag{6.2}$$

$$+ \sum_{\substack{b>k,\ wt_{kb}\lhd w\\ \ell(w)=\ell(wt_{kb})+1}} c_{k-1}(u,wt_{kb}) + \sum_{\substack{b>k,\ wt_{kb}\lhd w\\ \ell(w)=\ell(wt_{kb})-2(b-k)+1}} q_{kb}\,c_{k-1}(u,wt_{kb}) \tag{6.3}$$

$$- \sum_{\substack{a<k,\ wt_{ak}\lhd w\\ \ell(w)=\ell(wt_{ak})+1}} c_{k-1}(u,wt_{ak}) - \sum_{\substack{a<k,\ wt_{ak}\lhd w\\ \ell(w)=\ell(wt_{ak})-2(k-a)+1}} q_{ak}\,c_{k-1}(u,wt_{ak}) \tag{6.4}$$

$$+ q_{k-1}\,c_{k-2}(u,w), \tag{6.5}$$

where we used that $\mathrm{sign}(b-k)=1$ for $b>k$ and $\mathrm{sign}(a-k)=-1$ for $a<k$, and that a summand in (6.3)–(6.4) is present precisely when $u \overset{k-1}{\rightsquigarrow} wt_{kb}$, respectively $u \overset{k-1}{\rightsquigarrow} wt_{ak}$ (otherwise $c_{k-1}=0$). We must show this equals $c_k(u,w)$. We distinguish three cases according to whether $u(k)=w(k)$ and, when $u(k)\neq w(k)$, whether $u \overset{k}{\rightsquigarrow} w$; by Lemma 6.10 these cases are exhaustive and, in particular, the combination $u(k)\neq w(k)$ with both $u \overset{k}{\rightsquigarrow} w$ and $u \overset{k-1}{\rightsquigarrow} w$ never occurs.

Case 1: $u(k)=w(k)$. By Lemma 6.10(i) we have $u \overset{k}{\rightsquigarrow} w$ if and only if $u \overset{k-1}{\rightsquigarrow} w$; suppose first that both hold. Since $u(k)=w(k)$,

$$\mathrm{fix}_k(u,w) = \mathrm{fix}_{k-1}(u,w)\cup\{u(k)\}, \qquad q_k(u,w)=q_{k-1}(u,w),$$

the second equality holding because no reflection with first index k occurs. The diagonal contribution (6.2) then gives

$$(y_{u(k)}-z_1)\,c_{k-1}(u,w) = q_{k-1}(u,w)\,(y_{u(k)}-z_1)\,\mathbf{E}(y_{\mathrm{fix}_{k-1}(u,w)};z) = q_k(u,w)\,\mathbf{E}(y_{\mathrm{fix}_k(u,w)};z) = c_k(u,w),$$

by adjoining the variable $y_{u(k)}$ as in Lemma 6.15. The remaining contributions are the reflection sums (6.3), (6.4) and the quantum term $q_{k-1}c_{k-2}(u,w)$ of (6.5), which sum to 0 by Lemma 6.11. Hence the total is $c_k(u,w)$, as required. If instead $u(k)=w(k)$ but $u \overset{k}{\not\rightsquigarrow} w$ (equivalently $u \overset{k-1}{\not\rightsquigarrow} w$), then $c_k(u,w)=0$ and the diagonal term vanishes, while the reflection sums and quantum term again cancel by Lemma 6.11; the total is $0=c_k(u,w)$.

Case 2: $u \overset{k}{\rightsquigarrow} w$ *but* $u \overset{k-1}{\not\rightsquigarrow} w$. By Lemma 6.9 there is a unique $b>k$ with $wt_{kb}\lhd w$ and $u \overset{k-1}{\rightsquigarrow} wt_{kb}$; set $v=wt_{kb}$. This is exactly the unique nonvanishing summand of (6.3), and by Lemma 6.9 the sums (6.4) vanish, as does the diagonal (6.2) and the quantum term (6.5) (since $u \overset{k-1}{\not\rightsquigarrow} w$ and $u \overset{k-2}{\not\rightsquigarrow} w$). Moreover $\mathrm{fix}_k(u,w)=\mathrm{fix}_{k-1}(u,v)$, so

$$\mathbf{E}(y_{\mathrm{fix}_k(u,w)};z) = \mathbf{E}(y_{\mathrm{fix}_{k-1}(u,v)};z).$$

If $\ell(w) = \ell(v) + 1$ then the reflection t_{kb} is length-increasing, so by Definition 3.1 $q_k(u,w) = q_{k-1}(u,v)$ and the contribution is $c_{k-1}(u,v) = c_k(u,w)$. If instead $\ell(w) = \ell(v) - 2(b-k) + 1$ then t_{kb} is length-decreasing, Definition 3.1 gives $q_k(u,w) = q_{k-1}(u,v)\, q_{kb}$, and the contribution is $q_{kb}\, c_{k-1}(u,v) = c_k(u,w)$. In both cases the coefficient equals $c_k(u,w)$, as required. This is the step at which the quantum drop terms of Lemma 6.1 and the recursive definition of $q_k(u,w)$ interlock.

Case 3: $u \overset{k}{\not\rightarrow} w$ (with $u(k) \neq w(k)$). Here $c_k(u,w) = 0$, so we must show the coefficient (6.2)–(6.5) vanishes. This is exactly the assertion of Lemma 6.14. (The subcase $u(k) = w(k)$ with $u \overset{k}{\not\rightarrow} w$ was already treated at the end of Case 1.)

In all cases $[\widetilde{\mathfrak{S}}_w](\widetilde{\mathfrak{S}}_u\, \widetilde{E}_{k,k}) = c_k(u,w)$, completing the induction and the proof. □

7. Additional formulas

Definition 7.1. Consider a permutation $d_2[k]$ satisfying

$$\mathfrak{c}(d_2[k]) = \overbrace{(2,2,\ldots,2)}^{k \text{ times}}.$$

The double Schubert polynomial $\mathfrak{S}_{d_2[k]}(x;y)$ is a product of two factorial elementary symmetric polynomials, specifically:

$$\mathfrak{S}_{d_2[k]}(x;y) = E_k(x;y_1)E_k(x;y_2).$$

Therefore, the Pieri formula in [7] easily transfers to being able to multiply by $\mathfrak{S}_{d_2[k]}(x;y)$.

For quantum double Schubert polynomials, $\widetilde{\mathfrak{S}}_{d_2[k]}(x;y)$ is *not* a product of factorial quantum elementary polynomials. For example:

$$\widetilde{\mathfrak{S}}_{d_2[1]}(x;y) = \widetilde{E}_{1,2}(x;y)\widetilde{E}_{1,1}(x;y) - E^q_{2,2}(x;y)$$

$$\widetilde{\mathfrak{S}}_{d_2[2]}(x;y) = E^q_{2,3}(x;y)E^q_{2,2}(x;y) - E^q_{3,3}(x;y)E^q_{1,2}(x;y)$$

For abbreviation purposes, we denote this polynomial by $D^q_k(x;y)$.

We define a relation $\overset{k}{\rightrightarrows}$ on permutations as follows. We say that $u \overset{k}{\rightrightarrows} w$ if and only if there exists a $v \in S_\infty$ such that $u \overset{k}{\rightsquigarrow} v$ and $v \overset{k}{\rightsquigarrow} w$ satisfying the additional condition that if $u^{-1}v$ is a product of disjoint cycles $c_1, \ldots, c_m$ and $v^{-1}w$ is a product of disjoint cycles $d_1, \ldots, d_n$, then for each d_i there exists a c_j such that $d_i \cap c_j = \{\max(d_i)\} = \{\max(c_j)\}$. Call the set of all such v $\mathrm{pivot}_k(u,w)$.

Then

Theorem 7.2 (Doubled quantum Pieri formula)**.** *We have that the product $\widetilde{\mathfrak{S}}_u(x;y)D^q_k(x;z)$ is equal to*

$$\sum_{u \overset{k}{\rightrightarrows} w} \sum_{v \in \mathrm{pivot}_k(u,w)} q_k(u,v)q_k(v,w) \prod_{a \in \mathrm{fix}_k(u,v)} (y_a - z_1) \prod_{b \in \mathrm{fix}_k(v,w)} (y_b - z_2)\, \widetilde{\mathfrak{S}}_w(x;y).$$

Corollary 7.3 (Iterated quantum Pieri formula)**.** *For all $u \in S_\infty$ and $k \geq 1$,*

$$\widetilde{\mathfrak{S}}_u(x;y)\widetilde{E}_{k,k}(x;z_1)\widetilde{E}_{k,k}(x;z_2) = \sum_{u \overset{k}{\rightsquigarrow} v \overset{k}{\rightsquigarrow} w} q_k(u,v)q_k(v,w) \prod_{a \in \mathrm{fix}_k(u,v)} (y_a - z_1) \prod_{b \in \mathrm{fix}_k(v,w)} (y_b - z_2)\, \widetilde{\mathfrak{S}}_w(x;y).$$

Proof. Apply Theorem 3.2 to multiply $\widetilde{\mathfrak{S}}_u(x;y)$ by $\widetilde{E}_{k,k}(x;z_1)$. For each resulting term indexed by v, apply the same theorem with the z-alphabet replaced by z_2. The coefficient factors multiply, and the two successive relations are exactly $u \overset{k}{\rightsquigarrow} v \overset{k}{\rightsquigarrow} w$. □

8. Peterson-Woodward comparison formula

Proposition 8.1 ([3, Proposition 2.10])**.** *Let W_P be a parabolic subgroup of W. Then for any $\lambda_P \in Q^\vee/Q^\vee_P$, we have that*

$$c^{w,\lambda_P}_{u,v}(y;y) = c^{w\omega_P\omega_{P'},\lambda_B}_{u,v}(y;y),$$

where ω_P is the longest element in W_P, $\omega_{P'}$ is the longest element in $W_{P'}$, where P' is the parabolic subgroup generated by $\{s_\alpha \mid \alpha \in \Delta_P, \langle \alpha, \lambda_B \rangle = 0\}$, and λ_B is the Peterson-Woodward lift of λ_P.

Corollary 8.2. *The Peterson-Woodward comparison formula holds for $c^w_{u,v}(y;z)$.*

Proof. We have that

$$c_{u,v}^{w,\lambda_P}(y;z) = \sum_{v_1 \cdot v_2 = v} c_{u,v_2}^{w,\lambda_P}(y;y)\, \mathfrak{S}_{v_1}(y;z)$$

By definition of a parabolic subgroup, $v_2 \in W_P$ as well. Thus, this is equal to

$$\sum_{v_1 \cdot v_2 = v} c_{u,v_2}^{w\omega_P\omega_{P'},\lambda_B}(y;y)\, \mathfrak{S}_{v_1}(y;z) = c_{u,v}^{w\omega_P\omega_{P'},\lambda_B}(y;z).$$

□

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